

Open Play: Data Understanding

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This document contains the initial exploration of the Open Play dataset, but focuses on getting an analysis-ready intake dataset.

This document explores the clean version of the data set Open Play. This is to help me get a better understanding of the data I will be working with for my MSc dissertation. The cleaning process consisted of the following:

- Steam: -Removed idle games or non games (SteamVR, soundpad, etc.), - Added approximate session times - Scaled sessions over an hour - Boolean indicator for new sessions - Added number of hours sessions lasted - Calculated metrics for each session/game id - Created quality metrics (separate file containing information about the scaled instances)
- Xbox: - Merge adjacent sessions that are within a threshold (60 seconds) - Remove non-game sessions (shopping/videos) - Remove suspiciously long sessions (more than 6 hours) - Merges adjacent sessions by splitting into lots of atomic sessions and merging
- Nintendo: - Filter duplicates - Merge adjacent sessions with a minute tolerance
- Android: simple renaming
- iOS: simple renaming - Excludes entries where total gaming for a day is more than 16 hours
- intake: - Self-reported hours > 50, NA - Remove duplicates of linked_platforms
- daily: - Remove daily duplicates of survey - Recode erroneous wave codings by their start date - Ignore “false starts” of game data (outside date range)
- bikweekly: - Remove daily duplicates of survey - Recode erroneous wave codings by their start date - Ignore “false starts” of game data (outside date range)
- timeuse: assign quality flags to diaries - No mention of sleep or personal care - No sleep - More than 90 minutes unaccounted for - Less than 5 activities logged during a day
- simon: - Only include non-pilot-study participants (weren’t in initial pre-trial) who qualified at intake

I will fetch the cleaned datasets:

```
# clean most likely to use columns
steam_visibility <- steam_visibility |>
  filter(platform == "steam") |>
  mutate(
    timestamp = as_datetime(timestamp, tz = "UTC"),
    is_public = event == "steamDataPublic"
  ) |>
  select(pid, timestamp, event, is_public)
```

```

# import raw data for calculating how many rows were excluded
xbox_valid_sessions <- xbox_raw |>
  merge_adjacent_sessions(tol_sec = 60) |>
  # filter non-foreground games
  filter(
    duration >= 1,
    pid %in% intake$pid,
    title_placement == "Full", # in foreground
    !genres %in% c("Shopping", "Video", "Entertainment", "Utilities & tools"),
    tolower(title_id) != "e1924c10-9c91-4652-81b9-8ca0670e77a7" # home screen
  )

nintendo_valid_sessions <- nintendo_raw |>
  distinct() |>
  filter(
    pid %in% intake$pid,
    duration >= 1
  ) |>
  merge_adjacent_sessions(tol_sec = 60)

```

Now I shall select the qualified participants, as I know I will have more data to analyse. These participants are known to have completed at least one (biweekly) survey.

```

# merge with the local time zone offsets from intake
# Note: We keep country and local_timezone for DST-aware conversion
tz_map <- intake |>
  mutate(
    pid,
    country,
    local_timezone,
    .keep = "none"
  ) |>
  distinct(pid, .keep_all = TRUE)

# aggregate data at each level of granularity (session, hourly, daily)
# session level (Nintendo and Xbox)
session_telemetry <- bind_rows(
  xbox |> mutate(platform = "Xbox"),
  nintendo |> mutate(platform = "Nintendo")
) |>
  left_join(tz_map, by = "pid") |>

```

```

filter(!is.na(local_timezone)) |>
mutate(
  # Calculate DST-aware offset for each timestamp
  offset_start = get_dst_offset(session_start, country, local_timezone),
  offset_end = get_dst_offset(session_end, country, local_timezone),
  # Add offset to get local time values (keeping UTC label for compatibility)
  start_local = session_start + offset_start,
  end_local = session_end + offset_end,
  duration_min = as.numeric(difftime(
    session_end,
    session_start,
    units = "mins"
  ))
) |>
filter(
  !is.na(session_start),
  !is.na(session_end),
  session_end > session_start,
  duration_min >= 1
)

# Hourly (Nintendo and Xbox)
# expand sessions into local-hour bins, compute overlap minutes, and add UTC hour
hourly_from_sessions <- session_telemetry |>
  filter(!is.na(start_local), !is.na(end_local)) |>
  mutate(
    h0_local = floor_date(start_local, "hour"),
    h1_local = floor_date(end_local - seconds(1), "hour"),
    n_hours = as.integer(difftime(h1_local, h0_local, units = "hours")) + 1
  ) |>
  filter(!is.na(n_hours), n_hours > 0) |>
  tidyr::uncount(n_hours, .remove = FALSE, .id = "k") |>
  mutate(
    hour_start_local = h0_local + hours(k - 1),
    minutes = pmax(
      0,
      as.numeric(difftime(
        pmin(end_local, hour_start_local + hours(1)),
        pmax(start_local, hour_start_local),
        units = "mins"
      ))
    ),
  ),

```

```

    # Convert back to UTC (this preserves the instant, just changes label)
    hour_start_utc = with_tz(hour_start_local, tzone = "UTC")
  ) |>
  select(pid, platform, title_id, hour_start_local, hour_start_utc, minutes) |>
  group_by(pid, platform, title_id, hour_start_local, hour_start_utc) |>
  summarise(minutes = sum(minutes, na.rm = TRUE), .groups = "drop")

hourly_from_steam <- steam |>
  mutate(
    datetime_hour_start = ymd_hms(
      paste0(date, " ", sprintf("%02d", hour), ":00:00"),
      tz = "UTC"
    )
  ) |>
  select(pid, title_id, datetime_hour_start, minutes) |>
  left_join(tz_map, by = "pid") |>
  filter(!is.na(local_timezone)) |>
  mutate(
    platform = "Steam",
    hour_start_utc = datetime_hour_start,
    offset = get_dst_offset(datetime_hour_start, country, local_timezone),
    # Note: These are local time values with UTC labels due to R's POSIXct limitation
    hour_start_local = datetime_hour_start + offset
  ) |>
  select(pid, platform, title_id, hour_start_local, hour_start_utc, minutes) |>
  group_by(pid, platform, title_id, hour_start_local, hour_start_utc) |>
  summarise(minutes = sum(minutes, na.rm = TRUE), .groups = "drop")

hourly_telemetry <- bind_rows(hourly_from_sessions, hourly_from_steam)

# Daily (Nintendo + Xbox + Steam; collapse hourly to days)
daily_telemetry <- hourly_telemetry |>
  mutate(
    day_local = as.Date(hour_start_local),
  ) |>
  group_by(pid, platform, day_local) |>
  summarise(minutes = sum(minutes, na.rm = TRUE), .groups = "drop") |>
  bind_rows(ios) |>
  bind_rows(android)

# Weekly
weekly_all <- daily_telemetry |>

```

```

mutate(
  week = floor_date(day_local, "week")
) |>
group_by(pid, platform, week) |>
summarise(minutes = sum(minutes, na.rm = TRUE), .groups = "drop")

telemetry_spans <- daily_telemetry |>
group_by(pid, platform) |>
summarise(
  telemetry_start = min(day_local, na.rm = TRUE),
  telemetry_end = max(day_local, na.rm = TRUE) + hours(1), # end of last hour bin
  week = floor_date(telemetry_end, "week"),
  n_weeks = as.integer(difftime(
    telemetry_end,
    telemetry_start,
    units = "weeks"
  )) +
  1,
  .groups = "drop"
)

```

```

# participants
n_baseline <- nrow(intake[intake$qualified, ])
# participants in surveys
n_participants <- sum(unique(c(daily$pid, biweekly$pid)) %in% intake$pid)
n_participants_biweekly <- sum(unique(c(biweekly$pid)) %in% intake$pid)
n_participants_daily <- sum(unique(c(daily$pid)) %in% intake$pid)

participant_age <- (function(x) {
  sprintf(
    "%s years old (10th percentile: %s, 90th percentile: %s)",
    print_num(median(x), .1),
    print_num(quantile(x, 0.10), .1),
    print_num(quantile(x, 0.90), .1)
  )
})(na.omit(intake$age[intake$qualified]))

n_daily <- nrow(daily)
n_biweekly <- nrow(biweekly)
n_sessions <- (nrow(nintendo) + nrow(xbox) + nrow(steam))
n_hours <- (
  sum(nintendo$duration, na.rm = TRUE) +

```

```

sum(xbox$duration, na.rm = TRUE) +
sum(steam$minutes, na.rm = TRUE) +
sum(ios$minutes, na.rm = TRUE) +
sum(android$minutes, na.rm = TRUE)) /
60

n_platforms <- table(
  str_count(
    intake$linked_platforms[!is.na(intake$linked_platforms)],
    ", "
  ) +
  1
)

n_hours <- sum(daily_telemetry$minutes / 60)
n_users <- length(unique(daily_telemetry$pid))
n_games <- length(unique(hourly_telemetry$title_id))
n_months <- round(max(telemetry_spans$n_weeks) / 4.2)
n_surveys <- nrow(daily) + nrow(biweekly)
n_simon <- nrow(simon)

prop_third_party <- (1 - median(intake$nintendo_first_party_prop, na.rm = T)) * 100
prop_failed_daily_attn <- (sum(daily_raw$bpnsfs_failed_att_check) / nrow(daily_raw)) * 100
prop_failed_biweekly_attn <- (sum(biweekly_raw$bangs_failed_att_check) / nrow(biweekly_raw))

# no. invalid sessions (shopping, videos, etc.)
prop_invalid_xbox <- (1 - (nrow(xbox_valid_sessions) / nrow(xbox_raw))) * 100
prop_invalid_nintendo <- (1 - (nrow(nintendo_valid_sessions) / nrow(nintendo_raw))) * 100

# no. suspicious sessions (1 - proportion of clean session after removing long hour sessions)
prop_suspicious_xbox <- (1 - (nrow(xbox) / nrow(xbox_valid_sessions))) * 100
prop_suspicious_nintendo <- (1 - (nrow(nintendo) / nrow(nintendo_valid_sessions))) * 100

prop_dropped_timeuse <- (1 - (nrow(time_use) / nrow(time_use_raw))) * 100

n_timeuse_entries_per_day <- time_use |>
  group_by(pid, date) |>
  summarise(n_activites = n()) |>
  summarise(median = median(n_activites, na.rm = TRUE)) |>
  summarise(median = median(median, na.rm = TRUE)) |>
  pull(median)

```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by pid and date.
i Output is grouped by pid.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(pid, date))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
# This was wrong in the paper? see code chunk below
platforms_count <- str_count(
  replace_na(intake$linked_platforms[intake$qualified], ""),
  ",",
) + 1 # single platform won't have a ','
linked_platforms_raw_qual <- intake$linked_platforms[intake$qualified]
platforms_count_fix <- if_else(
  is.na(linked_platforms_raw_qual) | trimws(linked_platforms_raw_qual) == "",
  0,
  str_count(replace_na(linked_platforms_raw_qual, ""), ",") + 1
)
n_one_platform <- sum(platforms_count == 1, na.rm = TRUE)
n_two_platforms <- sum(platforms_count == 2, na.rm = TRUE)
n_three_platforms <- sum(platforms_count >= 3, na.rm = TRUE)

# Steam quality metrics
steam_qc <- read_csv(
  "data/open-play/qc/steam_quality_metrics.csv",
  show_col_types = FALSE
)
prop_steam_hours_exceeding <- steam_qc$prop_hours_exceeding * 100
prop_steam_segments_scaled <- steam_qc$prop_segments_scaled * 100

mean_minutes_reduced <- steam_qc$mean_minutes_reduced
median_minutes_reduced <- steam_qc$median_minutes_reduced
```

There are 2544 qualified participants and 1978 took part in the surveys. Out of the 1978, 1840 completed the biweekly surveys, and 1208 did the daily surveys.

For future reference, the variable `eligible_platforms` contains all platforms that can be used for telemetry.

```
table(intake$eligible_platforms, useNA = 'ifany')
```

	Nintendo	Steam	Steam, Nintendo
	2015	3637	1490
	Xbox	Xbox, Nintendo	Xbox, Steam
	4402	1145	1504
Xbox, Steam, Nintendo		<NA>	
	1208	19968	

The people that did not link their video game services, but were eligible, were initially classified as “Elected not to link”. Additionally, the participants did end up recording their telemetry, as seen in Figure 1 of the Open Play paper. Therefore, this finding simply indicates the participants at the time elected to not link, but did in the end.

Intake Data

```
intake_participants <- intake |> filter(pid %in% unique(c(daily$pid, biweekly$pid)))
intake_qualified <- intake |> filter(qualified)
```

First, let’s get an idea of the participants that took part in the surveys (both daily and biweekly).

```
print(str(intake_participants, list.len = Inf, vec.len = 3))
```

```
spc_tbl_ [1,978 x 88] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ pid                : chr [1:1978] "p8391238483" "p4295136583" "p3638519235" .
 $ date               : POSIXct[1:1978], format: "2025-02-10 18:05:12" "2024-
11-29 15:22:54" ...
 $ qualified          : logi [1:1978] TRUE TRUE TRUE TRUE ...
 $ survey_duration    : num [1:1978] 588 1266 878 1847 ...
 $ country            : chr [1:1978] "US" "US" "US" ...
 $ geo_area           : chr [1:1978] "123" "606" "921" ...
 $ panel_source       : chr [1:1978] "PureProfile" "PureProfile" "PureProfile" .
 $ cohort             : chr [1:1978] "Platform Pureprofile" "Platform Pureprofil
 $ diaries_completed  : num [1:1978] 0 0 0 0 0 0 0 0 ...
 $ panels_completed   : num [1:1978] 0 0 0 0 0 0 0 0 ...
 $ local_timezone     : chr [1:1978] NA "GMT-0600" "GMT-0800" ...
 $ age                : num [1:1978] 33 25 24 30 29 27 25 28 ...
 $ gender             : chr [1:1978] "Man" "Man" "Man" ...
 $ height             : num [1:1978] NA 183 173 165 ...
 $ weight             : num [1:1978] NA 61.2 84.4 72.6 ...
```



```

$ ethnicity : chr [1:1978] "White" "White alone" "Two or More Races" .
$ edu_level : chr [1:1978] "Higher National Certificate, Higher National
graduate qualifications" "Bachelor's degree" "Bachelor's degree" ...
$ employment : chr [1:1978] "Working full-time" "Working full-
time" "Working full-time" ...
$ marital_status : chr [1:1978] "In a domestic partnership or civil union"
$ political_party : chr [1:1978] "Democratic Party" "Democratic Party" "Indep
$ self_reported_weekly_play : num [1:1978] NA NA NA NA NA NA NA NA ...
$ used_platforms : chr [1:1978] "Steam,Xbox,Nintendo Switch" "Steam,Playsta
$ eligible_platforms : chr [1:1978] "Xbox, Steam, Nintendo" "Xbox, Steam, Ninter
$ linked_platforms : chr [1:1978] NA "Nintendo" "Nintendo" ...
$ plays_xbox : logi [1:1978] TRUE TRUE FALSE FALSE ...
$ plays_steam : logi [1:1978] TRUE TRUE FALSE TRUE ...
$ plays_nintendo : logi [1:1978] TRUE TRUE TRUE FALSE ...
$ plays_ios : logi [1:1978] FALSE TRUE TRUE FALSE ...
$ plays_android : logi [1:1978] FALSE FALSE FALSE TRUE ...
$ plays_playstation : logi [1:1978] FALSE TRUE TRUE FALSE ...
$ prop_steam : num [1:1978] 41 14 NA 60 NA 50 NA NA ...
$ prop_xbox : num [1:1978] 29 17 NA NA 86 NA NA 50 ...
$ prop_playstation : num [1:1978] NA 7 70 NA NA 40 NA NA ...
$ prop_nintendo : num [1:1978] 30 51 20 NA NA NA 100 NA ...
$ prop_ios : num [1:1978] NA 11 10 NA 14 10 NA NA ...
$ prop_android : num [1:1978] NA NA NA 40 NA NA NA 50 ...
$ prop_other : num [1:1978] NA NA NA NA NA NA NA NA ...
$ nintendo_uses_family : chr [1:1978] "No" "No" "No" ...
$ android_uses_family : chr [1:1978] NA NA NA ...
$ ios_uses_family : chr [1:1978] NA "No" "No" ...
$ steam_uses_family : chr [1:1978] NA "No" NA ...
$ xbox_uses_family : chr [1:1978] NA "No" NA ...
$ total_recent_play_accounted_for : num [1:1978] 0 0 20 0 0 0 0 0 ...
$ total_play_possible_to_collect : num [1:1978] 0 51 20 60 0 50 100 0 ...
$ neuro_identify : chr [1:1978] "No" "No" "No" ...
$ neuro_diagnosed : chr [1:1978] NA NA NA ...
$ neuro_diag_asd : chr [1:1978] NA NA NA ...
$ neuro_diag_adhd : chr [1:1978] NA NA NA ...
$ neuro_diag_dyslexia : chr [1:1978] NA NA NA ...
$ neuro_diag_dyspraxia : chr [1:1978] NA NA NA ...
$ neuro_diag_dyscalculia : chr [1:1978] NA NA NA ...
$ neuro_diag_tourette : chr [1:1978] NA NA NA ...
$ neuro_diag OCD : chr [1:1978] NA NA NA ...
$ neuro_diag_spd : chr [1:1978] NA NA NA ...
$ neuro_diag_nvld : chr [1:1978] NA NA NA ...
$ neuro_diag_specific_ld : chr [1:1978] NA NA NA ...

```

```

$ neuro_diag_prefer_not_say      : chr [1:1978] NA NA NA ...
$ neuro_diag_other               : chr [1:1978] NA NA NA ...
$ neuro_iden_asd                 : chr [1:1978] NA NA NA ...
$ neuro_iden_adhd                : chr [1:1978] NA NA NA ...
$ neuro_iden_dyslexia            : chr [1:1978] NA NA NA ...
$ neuro_iden_dyspraxia           : chr [1:1978] NA NA NA ...
$ neuro_iden_dyscalculia         : chr [1:1978] NA NA NA ...
$ neuro_iden_tourette            : chr [1:1978] NA NA NA ...
$ neuro_iden OCD                 : chr [1:1978] NA NA NA ...
$ neuro_iden_spd                 : chr [1:1978] NA NA NA ...
$ neuro_iden_nvld                : chr [1:1978] NA NA NA ...
$ neuro_iden_specific_ld         : chr [1:1978] NA NA NA ...
$ neuro_iden_prefer_not_say      : chr [1:1978] NA NA NA ...
$ neuro_iden_other               : chr [1:1978] NA NA NA ...
$ care_children                  : chr [1:1978] NA NA NA ...
$ care_elderly                   : chr [1:1978] NA NA NA ...
$ care_disability                : chr [1:1978] NA NA NA ...
$ care_none                      : chr [1:1978] NA "No, I do not have additional caregiving
$ care_prefer_not_say            : chr [1:1978] "Prefer not to say" NA NA ...
$ care_other                     : chr [1:1978] NA NA NA ...
$ dependents                     : chr [1:1978] NA NA NA ...
$ life_sat                       : num [1:1978] NA NA NA NA NA NA NA NA ...
$ wemwbs_1                       : chr [1:1978] NA NA NA ...
$ wemwbs_2                       : chr [1:1978] NA NA NA ...
$ wemwbs_3                       : chr [1:1978] NA NA NA ...
$ wemwbs_4                       : chr [1:1978] NA NA NA ...
$ wemwbs_5                       : chr [1:1978] NA NA NA ...
$ wemwbs_6                       : chr [1:1978] NA NA NA ...
$ wemwbs_7                       : chr [1:1978] NA NA NA ...
$ affective_valence              : num [1:1978] NA NA NA NA NA NA NA NA ...
$ nintendo_account_creation_date : POSIXct[1:1978], format: NA "2017-12-
26 18:58:32" ...
$ nintendo_first_party_prop      : num [1:1978] NA 0.742 0.228 NA ...
- attr(*, "spec")=
.. cols(
..   pid = col_character(),
..   date = col_datetime(format = ""),
..   qualified = col_logical(),
..   survey_duration = col_double(),
..   country = col_character(),
..   geo_area = col_character(),
..   panel_source = col_character(),
..   cohort = col_character(),

```

```

.. diaries_completed = col_double(),
.. panels_completed = col_double(),
.. local_timezone = col_character(),
.. age = col_double(),
.. gender = col_character(),
.. height = col_double(),
.. weight = col_double(),
.. ethnicity = col_character(),
.. edu_level = col_character(),
.. employment = col_character(),
.. marital_status = col_character(),
.. political_party = col_character(),
.. self_reported_weekly_play = col_double(),
.. used_platforms = col_character(),
.. eligible_platforms = col_character(),
.. linked_platforms = col_character(),
.. plays_xbox = col_logical(),
.. plays_steam = col_logical(),
.. plays_nintendo = col_logical(),
.. plays_ios = col_logical(),
.. plays_android = col_logical(),
.. plays_playstation = col_logical(),
.. prop_steam = col_double(),
.. prop_xbox = col_double(),
.. prop_playstation = col_double(),
.. prop_nintendo = col_double(),
.. prop_ios = col_double(),
.. prop_android = col_double(),
.. prop_other = col_double(),
.. nintendo_uses_family = col_character(),
.. android_uses_family = col_character(),
.. ios_uses_family = col_character(),
.. steam_uses_family = col_character(),
.. xbox_uses_family = col_character(),
.. total_recent_play_accounted_for = col_double(),
.. total_play_possible_to_collect = col_double(),
.. neuro_identify = col_character(),
.. neuro_diagnosed = col_character(),
.. neuro_diag_asd = col_character(),
.. neuro_diag_adhd = col_character(),
.. neuro_diag_dyslexia = col_character(),
.. neuro_diag_dyspraxia = col_character(),
.. neuro_diag_dyscalculia = col_character(),

```

```

.. neuro_diag_tourette = col_character(),
.. neuro_diag OCD = col_character(),
.. neuro_diag_spd = col_character(),
.. neuro_diag_nvld = col_character(),
.. neuro_diag_specific_ld = col_character(),
.. neuro_diag_prefer_not_say = col_character(),
.. neuro_diag_other = col_character(),
.. neuro_iden_asd = col_character(),
.. neuro_iden_adhd = col_character(),
.. neuro_iden_dyslexia = col_character(),
.. neuro_iden_dyspraxia = col_character(),
.. neuro_iden_dyscalculia = col_character(),
.. neuro_iden_tourette = col_character(),
.. neuro_iden_OCD = col_character(),
.. neuro_iden_spd = col_character(),
.. neuro_iden_nvld = col_character(),
.. neuro_iden_specific_ld = col_character(),
.. neuro_iden_prefer_not_say = col_character(),
.. neuro_iden_other = col_character(),
.. care_children = col_character(),
.. care_elderly = col_character(),
.. care_disability = col_character(),
.. care_none = col_character(),
.. care_prefer_not_say = col_character(),
.. care_other = col_character(),
.. dependents = col_character(),
.. life_sat = col_double(),
.. wemwbs_1 = col_character(),
.. wemwbs_2 = col_character(),
.. wemwbs_3 = col_character(),
.. wemwbs_4 = col_character(),
.. wemwbs_5 = col_character(),
.. wemwbs_6 = col_character(),
.. wemwbs_7 = col_character(),
.. affective_valence = col_double(),
.. nintendo_account_creation_date = col_datetime(format = ""),
.. nintendo_first_party_prop = col_double()
.. )
- attr(*, "problems")=<externalptr>
NULL

```

```
print(summary(intake_participants))
```

pid	date	qualified	
Length:1978	Min. :2024-09-26 19:20:41	Mode:logical	
Class :character	1st Qu.:2025-02-27 19:56:08	TRUE:1978	
Mode :character	Median :2025-03-01 07:19:11		
	Mean :2025-03-23 15:55:37		
	3rd Qu.:2025-05-29 03:28:03		
	Max. :2025-06-09 09:05:44		
survey_duration	country	geo_area	panel_source
Min. : 46.0	Length:1978	Length:1978	Length:1978
1st Qu.: 393.2	Class :character	Class :character	Class :character
Median : 588.0	Mode :character	Mode :character	Mode :character
Mean : 779.2			
3rd Qu.: 928.8			
Max. :13428.0			
cohort	diaries_completed	panels_completed	local_timezone
Length:1978	Min. : 0.000	Min. :0.000	Length:1978
Class :character	1st Qu.: 0.000	1st Qu.:1.000	Class :character
Mode :character	Median : 0.000	Median :4.000	Mode :character
	Mean : 8.304	Mean :3.312	
	3rd Qu.:21.000	3rd Qu.:6.000	
	Max. :30.000	Max. :6.000	
age	gender	height	weight
Min. :18.00	Length:1978	Min. :143.5	Min. : 4.54
1st Qu.:23.00	Class :character	1st Qu.:167.6	1st Qu.: 65.77
Median :26.00	Mode :character	Median :175.3	Median : 79.83
Mean :26.72		Mean :173.9	Mean : 83.44
3rd Qu.:30.00		3rd Qu.:180.3	3rd Qu.: 95.25
Max. :40.00		Max. :457.2	Max. :226.80
		NA's :138	NA's :137
ethnicity	edu_level	employment	marital_status
Length:1978	Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

political_party	self_reported_weekly_play	used_platforms
Length:1978	Min. : 30	Length:1978
Class :character	1st Qu.: 600	Class :character
Mode :character	Median : 900	Mode :character
	Mean :1083	
	3rd Qu.:1500	
	Max. :3000	
	NA's :499	
eligible_platforms	linked_platforms	plays_xbox
Length:1978	Length:1978	Mode :logical
Class :character	Class :character	FALSE:1446
Mode :character	Mode :character	TRUE :532

plays_nintendo	plays_ios	plays_android	plays_playstation
Mode :logical	Mode :logical	Mode :logical	Mode :logical
FALSE:1235	FALSE:1432	FALSE:1512	FALSE:1520
TRUE :743	TRUE :546	TRUE :466	TRUE :458

prop_steam	prop_xbox	prop_playstation	prop_nintendo
Min. : 0.00	Min. : 0.00	Min. : 0.00	Min. : 0.00
1st Qu.: 50.00	1st Qu.: 12.00	1st Qu.: 7.00	1st Qu.: 7.00
Median : 80.00	Median : 40.00	Median :18.00	Median : 20.00
Mean : 72.23	Mean : 44.88	Mean :22.66	Mean : 30.21
3rd Qu.:100.00	3rd Qu.: 75.00	3rd Qu.:33.00	3rd Qu.: 41.00
Max. :100.00	Max. :100.00	Max. :91.00	Max. :100.00
NA's :339	NA's :1441	NA's :1525	NA's :1237
prop_ios	prop_android	prop_other	nintendo_uses_family
Min. : 0.00	Min. : 0.00	Min. : 0.00	Length:1978
1st Qu.: 7.00	1st Qu.: 9.00	1st Qu.:19.25	Class :character
Median :14.00	Median :15.00	Median :40.00	Mode :character
Mean :20.01	Mean :20.32	Mean :39.24	
3rd Qu.:28.00	3rd Qu.:28.75	3rd Qu.:60.00	
Max. :95.00	Max. :90.00	Max. :85.00	
NA's :1432	NA's :1512	NA's :1920	
android_uses_family	ios_uses_family	steam_uses_family	xbox_uses_family
Length:1978	Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character	Class :character

Mode :character Mode :character Mode :character Mode :character

total_recent_play_accounted_for	total_play_possible_to_collect
Min. : 0.00	Min. : 0.00
1st Qu.: 0.00	1st Qu.: 54.25
Median : 50.00	Median : 88.00
Mean : 47.64	Mean : 73.79
3rd Qu.: 95.00	3rd Qu.:100.00
Max. :100.00	Max. :100.00

neuro_identify	neuro_diagnosed	neuro_diag_asd	neuro_diag_adhd
Length:1978	Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

neuro_diag_dyslexia	neuro_diag_dyspraxia	neuro_diag_dyscalculia
Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

neuro_diag_tourette	neuro_diag OCD	neuro_diag_spd	neuro_diag_nvld
Length:1978	Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

neuro_diag_specific_ld	neuro_diag_prefer_not_say	neuro_diag_other
Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

neuro_iden_asd	neuro_iden_adhd	neuro_iden_dyslexia	neuro_iden_dyspraxia
Length:1978	Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

neuro_iden_dyscalculia	neuro_iden_tourette	neuro_iden OCD
Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

neuro_iden_spd	neuro_iden_nvld	neuro_iden_specific_ld
Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

neuro_iden_prefer_not_say	neuro_iden_other	Care_Children
Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

Care_elderly	Care_disability	Care_none	Care_prefer_not_say
Length:1978	Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

Care_other	dependents	life_sat	wemwbs_1
------------	------------	----------	----------

Length:1978	Length:1978	Min. : NA	Length:1978
Class :character	Class :character	1st Qu.: NA	Class :character
Mode :character	Mode :character	Median : NA	Mode :character
		Mean :NaN	
		3rd Qu.: NA	
		Max. : NA	
		NA's :1978	
wemwbs_2	wemwbs_3	wemwbs_4	wemwbs_5
Length:1978	Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

wemwbs_6	wemwbs_7	affective_valence
Length:1978	Length:1978	Min. : NA
Class :character	Class :character	1st Qu.: NA
Mode :character	Mode :character	Median : NA
		Mean :NaN
		3rd Qu.: NA
		Max. : NA
		NA's :1978

nintendo_account_creation_date	nintendo_first_party_prop
Min. :2016-02-17 09:28:57	Min. :0.0000
1st Qu.:2017-12-26 07:24:09	1st Qu.:0.3294
Median :2019-09-04 00:15:09	Median :0.6382
Mean :2019-10-27 13:53:08	Mean :0.5977
3rd Qu.:2020-12-19 15:36:01	3rd Qu.:0.8837
Max. :2025-06-04 09:06:37	Max. :1.0000
NA's :1382	NA's :1389

Missing Data Analysis

```
na_count <- unlist(colSums(is.na(intake_participants)))
na_prop <- unlist(round(colMeans(is.na(intake_participants)) * 100, 2))
intake_na <- data.table(variable = colnames(intake_participants), na_count, na_prop)

tot_na_count <- n_miss(intake_participants)
tot_na_pct <- pct_miss(intake_participants)
```

```
intake_na |> arrange(-na_count)
```

	variable	na_count	na_prop
	<char>	<num>	<num>
1:	neuro_iden_tourette	1978	100.00
2:	neuro_iden_nvld	1978	100.00
3:	life_sat	1978	100.00
4:	wemwbs_1	1978	100.00
5:	wemwbs_2	1978	100.00
6:	wemwbs_3	1978	100.00
7:	wemwbs_4	1978	100.00
8:	wemwbs_5	1978	100.00
9:	wemwbs_6	1978	100.00
10:	wemwbs_7	1978	100.00
11:	affective_valence	1978	100.00
12:	neuro_iden_dyscalculia	1977	99.95
13:	neuro_diag_nvld	1976	99.90
14:	neuro_diag_tourette	1975	99.85
15:	neuro_diag_prefer_not_say	1974	99.80
16:	neuro_iden_dyspraxia	1974	99.80
17:	neuro_iden_prefer_not_say	1973	99.75
18:	neuro_iden_other	1973	99.75
19:	neuro_iden_specific_ld	1972	99.70
20:	neuro_iden_spd	1967	99.44
21:	neuro_diag_dyscalculia	1965	99.34
22:	neuro_diag_specific_ld	1960	99.09
23:	care_prefer_not_say	1960	99.09
24:	neuro_iden_dyslexia	1959	99.04
25:	care_other	1958	98.99
26:	neuro_diag_dyspraxia	1955	98.84
27:	neuro_diag_spd	1955	98.84
28:	neuro_iden OCD	1950	98.58
29:	neuro_diag_dyslexia	1940	98.08
30:	neuro_diag_other	1933	97.72
31:	prop_other	1920	97.07
32:	neuro_diag_OCD	1912	96.66
33:	android_uses_family	1911	96.61
34:	care_elderly	1910	96.56
35:	care_disability	1907	96.41
36:	ios_uses_family	1833	92.67
37:	neuro_iden_asd	1829	92.47

38:	neuro_iden_adhd	1806	91.30
39:	neuro_diag_asd	1771	89.53
40:	care_children	1707	86.30
41:	neuro_diag_adhd	1647	83.27
42:	xbox_uses_family	1573	79.52
43:	dependents	1570	79.37
44:	prop_playstation	1525	77.10
45:	prop_android	1512	76.44
46:	prop_xbox	1441	72.85
47:	prop_ios	1432	72.40
48:	nintendo_first_party_prop	1389	70.22
49:	nintendo_account_creation_date	1382	69.87
50:	nintendo_uses_family	1240	62.69
51:	prop_nintendo	1237	62.54
52:	neuro_diagnosed	1187	60.01
53:	self_reported_weekly_play	499	25.23
54:	care_none	425	21.49
55:	steam_uses_family	383	19.36
56:	prop_steam	339	17.14
57:	height	138	6.98
58:	weight	137	6.93
59:	linked_platforms	122	6.17
60:	local_timezone	44	2.22
61:	ethnicity	44	2.22
62:	employment	1	0.05
63:	pid	0	0.00
64:	date	0	0.00
65:	qualified	0	0.00
66:	survey_duration	0	0.00
67:	country	0	0.00
68:	geo_area	0	0.00
69:	panel_source	0	0.00
70:	cohort	0	0.00
71:	diaries_completed	0	0.00
72:	panels_completed	0	0.00
73:	age	0	0.00
74:	gender	0	0.00
75:	edu_level	0	0.00
76:	marital_status	0	0.00
77:	political_party	0	0.00
78:	used_platforms	0	0.00
79:	eligible_platforms	0	0.00
80:	plays_xbox	0	0.00

81:	plays_steam	0	0.00
82:	plays_nintendo	0	0.00
83:	plays_ios	0	0.00
84:	plays_android	0	0.00
85:	plays_playstation	0	0.00
86:	total_recent_play_accounted_for	0	0.00
87:	total_play_possible_to_collect	0	0.00
88:	neuro_identify	0	0.00
	variable	na_count	na_prop
	<char>	<num>	<num>

Demographics

```
# exploring demographic variable missingness
demographic_vars <- c(
  "country",
  "geo_area",
  "age",
  "gender",
  "height",
  "weight",
  "ethnicity",
  "edu_level",
  "employment",
  "marital_status",
  "political_party",
  "dependents",
  "neuro_diagnosed",
  "neuro_identify"
)

intake_na |> filter(variable %in% demographic_vars) |> arrange(-na_count)
```

	variable	na_count	na_prop
	<char>	<num>	<num>
1:	dependents	1570	79.37
2:	neuro_diagnosed	1187	60.01
3:	height	138	6.98
4:	weight	137	6.93
5:	ethnicity	44	2.22

```

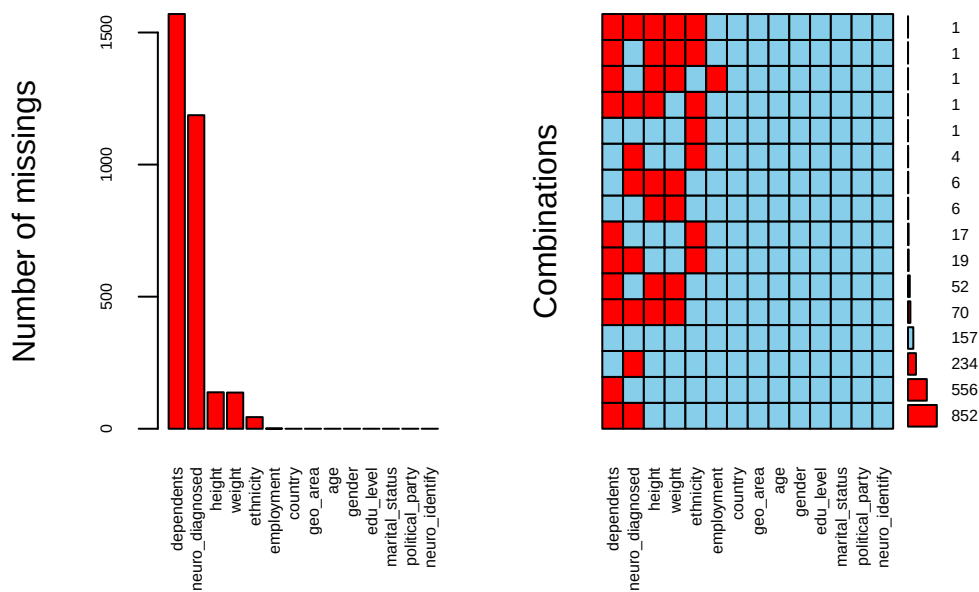
6:      employment      1    0.05
7:      country         0    0.00
8:      geo_area        0    0.00
9:      age             0    0.00
10:     gender          0    0.00
11:     edu_level       0    0.00
12:     marital_status  0    0.00
13:     political_party  0    0.00
14:     neuro_identify  0    0.00

```

```

aggr(
  intake_participants[, demographic_vars],
  sortVar = TRUE,
  numbers = TRUE,
  prop = FALSE,
  cex.lab = 1,
  cex.axis = 0.5,
  cex.numbers = 0.5
)

```



Variables sorted by number of missings:

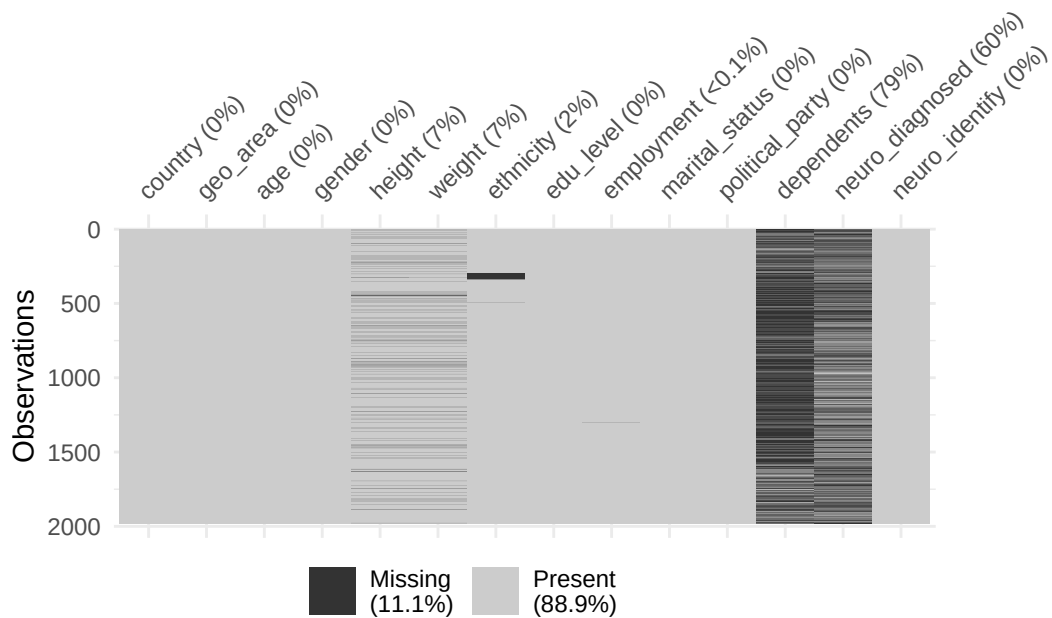
```

Variable Count
dependents    1570
neuro_diagnosed 1187

```

height	138
weight	137
ethnicity	44
employment	1
country	0
geo_area	0
age	0
gender	0
edu_level	0
marital_status	0
political_party	0
neuro_identify	0

```
vis_miss(intake_participants[, demographic_vars]) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 0, vjust = 0),
    margins = margin(r = 50, b = 5)
  )
```



We can see that the main variables missing data are dependents and neuro_diagnosed.

Cleaning

I will clean the variables that are not missing, but have lots of varied responses (country, geo_area, age, gender, edu_level, marital_status, political_party, neuro_identify).

Country

```
table(intake_participants$country, useNA = "ifany")
```

```
UK    US  
719 1259
```

There is nothing to be done with country.

Geographic Area of Residence

```
table(intake_participants$geo_area, useNA = "ifany")
```

```
010 012 014 015 018 019 020 021 022 023 024 027 028 029 030 031  
  1   1   2   1   2   1   1   7   1   3   1   6   2   3   2   3  
032 035 036 040 044 046 050 057 060 061 062 065 066 067 068 069  
  1   1   1   1   1   1   3   1   3   3   1   1   2   1   3   1  
070 073 076 077 078 079 080 081 082 084 085 088 100 103 104 106  
  4   2   1   1   1   1   2   2   1   1   1   4   3   4  10   1  
109 111 112 113 114 115 116 117 118 119 121 123 125 128 130 131  
  4   3   9   5   5   2   1   4   1   1   2   2   4   1   3   1  
132 138 140 142 144 145 146 147 150 151 152 157 158 160 163 166  
  2   1   1   5   1   1   3   2   2   1   7   1   2   3   1   2  
168 170 171 172 173 174 175 177 180 182 183 184 186 187 190 191  
  1   2   2   2   2   1   3   1   2   2   2   2   2   1   5   8  
193 194 196 197 198 201 207 208 210 211 212 215 217 219 220 221  
  1   2   1   1   1   3   6   1   4   2   4   2   1   2   4   2  
222 223 228 229 230 231 232 234 235 236 238 240 242 244 245 247  
  1   1   1   1   1   1   1   2   2   1   1   5   1   1   1   1  
250 251 252 253 257 260 263 265 271 272 273 274 275 276 277 278  
  2   1   1   1   1   1   1   1   1   2   1   2   7   6   2   3  
279 280 281 282 283 284 285 286 287 289 290 291 292 293 294 295  
  1   2   1   2   2   1   3   2   1   1   2   2   1   2   3   4  
296 298 300 301 302 303 304 305 306 307 308 309 310 312 313 314  
  2   1  10   5   1   5   1   1   2   3   1   1   2   2   2   1  
315 317 319 320 321 322 323 324 325 326 327 328 329 330 331 332  
  1   2   1   2   3  11   2   3   2   1   6  12   1   9   7   1  
334 335 336 337 338 339 341 342 344 346 347 349 350 351 352 356
```

5	3	3	4	6	7	2	5	5	2	2	4	4	2	3	2
358	362	368	370	371	372	373	374	376	378	379	380	381	383	385	386
2	1	1	6	3	1	1	2	2	1	2	1	3	1	3	3
388	389	391	393	395	398	400	401	402	403	404	405	407	409	410	412
2	2	1	1	1	2	1	2	1	2	3	3	1	1	1	1
426	427	430	431	432	436	439	440	441	442	443	444	446	448	449	450
1	2	1	2	2	2	1	4	6	5	2	1	4	2	2	3
451	452	453	454	456	460	461	462	465	467	469	474	477	478	479	480
1	3	4	5	4	2	1	4	2	2	1	1	1	1	3	3
481	482	483	486	488	489	490	492	493	494	495	498	500	501	510	515
6	4	2	3	2	1	2	2	2	1	2	1	3	1	1	1
522	524	527	530	531	532	535	537	541	544	547	548	549	553	554	556
1	1	1	1	3	3	1	1	2	1	2	1	2	4	7	1
558	562	563	565	567	571	581	582	585	587	591	597	598	5BR	600	601
1	1	1	1	1	1	1	1	1	1	1	1	2	1	6	6
603	604	605	606	608	609	611	612	613	614	616	617	618	619	622	627
1	5	5	12	2	1	1	2	1	1	1	1	3	1	2	1
628	630	631	633	640	641	646	647	651	652	658	668	670	672	674	675
1	5	3	2	4	4	1	1	1	1	2	1	2	3	1	2
678	680	681	683	684	685	689	700	701	703	704	705	706	707	708	710
1	1	2	1	1	2	1	2	2	1	2	3	2	1	1	1
711	712	713	720	721	727	730	731	736	740	749	750	751	752	754	756
2	1	1	1	1	4	1	7	1	2	1	8	5	1	2	1
759	760	761	762	765	767	770	773	774	775	777	778	780	781	782	783
2	3	6	4	1	1	11	7	5	10	1	2	1	1	11	2
785	786	787	789	790	791	794	799	800	801	802	806	808	809	813	814
7	5	7	1	1	1	1	5	5	2	3	1	3	3	1	2
832	834	836	837	840	841	843	844	846	850	851	852	853	856	857	868
1	1	3	1	10	3	2	2	2	4	2	6	7	1	5	1
871	890	891	894	895	900	901	902	903	905	907	908	911	912	913	916
2	3	9	3	2	8	1	5	1	1	1	1	1	1	3	1
917	918	919	920	921	922	923	924	925	926	927	928	930	931	932	933
10	2	2	3	6	2	5	1	9	5	2	3	3	1	3	6
934	935	937	939	940	941	945	946	950	951	952	953	954	956	957	958
4	5	1	2	2	5	7	1	1	4	2	5	2	4	3	3
959	960	967	970	973	974	977	980	981	982	983	984	985	986	989	990
2	1	2	4	7	2	1	6	5	5	2	2	5	5	2	2
991	992	993	996	998	AB11	AB16	ab52	AL1	b10	B13	B15	B16	B18	b20	B25
1	3	3	1	2	1	1	1	2	1	2	1	1	1	1	1
B29	B31	B45	B61	B63	B64	B68	B69	B71	B74	B77	b78	B98	BA1	BA2	BA21
1	1	1	3	1	1	1	1	2	1	1	1	1	1	2	1
BB4	BB5	BD23	Bd9	BH1	BH14	BH15	BH21	BH25	BH3	BH5	BL1	BL4	BL9	Bn1	BN1
1	1	2	1	1	1	1	1	1	1	1	2	1	1	1	3

BN10	BN16	BN17	BN2	bn23	BN23	BN3	BN41	BR3	BR6	BR7	BS1	Bs10	BS16	BS20	BS27
1	1	1	2	1	1	2	1	1	1	1	2	1	3	1	1
bs29	BS29	BS3	BS35	BS37	Bs4	BS5	BS7	BT23	BT28	BT34	BT36	BT47	BT48	BT61	bt8
1	1	2	1	1	1	3	1	1	1	1	1	1	2	1	1
BT9	CB	CB10	CB23	cb24	CB25	CF10	CF11	CF23	CF24	Cf31	CF35	CF38	Cf47	Cf62	CF62
1	1	1	1	1	1	1	2	2	3	1	1	1	1	1	1
CF64	CF72	CF82	CH1	CH7	CM1	CM14	Cm2	CM2	cm20	CM20	CM5	Cm6	CM7	CM8	co1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
CO5	CRO	CR7	CT12	CT14	CT2	cv1	CV10	CV2	CV23	CV31	CV5	CV9	CW1	CW11	DA!2
2	2	1	1	1	2	1	1	2	1	1	1	1	1	1	1
Da1	DA11	DA17	da5	da7	DD1	DD10	DD2	DD3	DD4	DD8	De12	de14	DE14	DE15	DE22
1	1	1	1	1	1	1	1	2	2	1	1	1	1	1	1
DE24	DE5	DE65	DE73	De74	DH3	DL14	dl17	DL3	DL5	DL9	DN14	Dn17	DN18	DN21	DN33
1	1	1	1	1	2	1	1	1	1	1	1	1	1	1	1
dn34	DN34	DN35	dn4	DN4	DN41	DT11	DT3	DT4	DT8	DT9	DY10	DY11	DY13	Dy2	dy5
1	1	2	1	1	1	1	1	1	1	1	1	2	1	1	1
DY9	E1	E14	E15	e16	E16	E17	e2	E3	E5	E6	EH11	EH12	EH13	EH14	EH16
1	2	2	1	1	2	1	1	1	1	1	1	1	1	2	1
EH17	EH20	EH25	EH28	Eh32	eh47	EH5	EH51	eh6	EH7	EH8	EX	EX16	EX2	EX3	EX32
1	1	1	1	1	1	1	1	1	2	1	1	1	1	1	1
Ex38	fk8	FK8	FY1	FY2	FY3	FY5	fy6	G11	G12	G14	G15	G21	G3	G31	G41
1	2	1	1	1	1	1	1	2	1	1	2	2	2	1	2
G49	G51	G64	G65	g67	g77	G82	GL1	GL2	g13	GL4	GL51	GL52	Gu1	GU11	gu12
1	1	1	1	1	1	1	1	2	1	1	2	1	1	1	1
GU12	GU2	GU24	GU27	GU3	gu30	gu35	gu47	GU5	Ha3	HA4	HA7	HA8	HA9	HD2	HD8
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
HP1	HP10	HP14	Hp2	HP21	HP23	HP7	HU6	HX2	hx5	ig11	IG11	IG3	IP14	Ip2	IP2
2	1	1	1	2	1	1	1	1	1	1	2	1	1	1	1
IP23	IP24	IV2	IV4	KA1	KA13	Ka3	ka4	KT1	kt22	KT22	KT8	KW14	ky10	Ky6	L12
1	1	1	1	1	1	1	1	2	1	1	1	1	1	1	1
L13	L17	l20	L20	L26	L3	L40	L6	L9	LA1	LA9	LE1	LE10	LE11	le15	LE16
1	1	1	2	1	1	1	1	1	2	1	2	1	1	1	1
LE17	LE18	LE2	LE8	LL12	LL21	Ll68	LN1	LN11	LN4	Ln5	ln6	LS11	LS12	LS13	LS14
2	1	2	1	1	1	1	1	2	1	1	1	1	1	2	2
LS17	LS27	LS28	LS4	LS6	LS7	LS8	LU4	LU7	m13	M15	M16	m23	M23	M25	m3
1	1	2	1	2	1	2	1	1	1	1	1	1	2	1	1
M3	M30	M32	M4	M40	M50	M54	ME1	ME10	ME12	ME14	ME15	ME3	MK14	Mk16	MK4
2	1	2	2	1	1	1	1	1	1	1	1	1	1	1	1
mk40	MK40	MK43	ML2	ML3	ml6	ML6	N1	N1 7	N11	n12	N12	N19	n21	N4	n7
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
N7	N8	NE1	NE10	Ne2	NE2	NE27	NE30	NE32	NE34	NE37	NE46	NE63	NE65	Ng10	NG11
1	1	2	1	1	1	1	1	1	1	1	1	1	1	1	1
NG12	NG13	NG15	Ng2	ng23	NG24	NG3	NG32	ng34	NG4	NG5	NG7	Ng8	NG8	NG9	NN1

1	1	1	1	1	1	2	1	1	3	1	2	1	2	3	1
NN2	NN7	NN9	NP16	NP19	NP22	NP26	NR1	NR2	NR26	NR3	NR30	NR31	NR5	NR8	NW10
1	1	1	1	2	1	1	1	2	1	1	1	2	1	1	1
NW8	OL2	OL4	OX1	ox11	OX11	Ox14	OX2	ox5	OX9	PA30	PA34	PE10	PE11	Pe2	PH1
1	1	1	3	1	1	1	1	1	1	1	1	1	1	1	1
ph10	PH2	PL1	PL19	PL2	P14	PL4	PL5	PL7	P01	P013	po14	P019	P02	P021	P022
1	1	1	1	1	1	1	2	1	1	1	1	1	1	1	1
P033	P04	P07	po8	P09	PR1	PR2	PR5	PR8	Rg1	RG1	RG10	RG2	rg22	RG22	RG27
1	1	1	1	2	1	1	1	1	1	4	2	1	1	1	1
rg4	rg40	rg41	RG5	RH1	rh19	RH2	RH6	RH8	RM15	RM8	S1	S11	S12	S13	S14
1	1	1	1	1	1	1	1	1	1	2	1	1	1	1	1
S2	S20	S26	S3	S41	s42	S6	s62	S62	S70	S71	S73	S75	S8	S9	sa1
2	1	1	1	1	1	2	1	1	1	1	1	1	2	1	1
SA1	Sa12	SA15	SA18	SA6	SA62	SA8	SE	se1	SE1	SE18	SE23	se28	SE3	SE6	SE9
3	1	1	1	1	1	1	1	1	3	2	2	1	1	1	1
SG1	SG14	SG4	SG7	SK11	SK13	SK4	SK8	SM2	SM6	sn1	SN1	SN16	SN7	S015	S016
1	1	1	1	1	1	2	1	1	1	1	1	1	1	1	1
So18	so2	S022	S023	S05	S051	SP4	SP6	SP8	SR3	SR4	SR5	SR7	SS1	SS14	Ss15
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
SS15	ST1	ST17	SW11	SW17	SW19	SW8	SW9	TA1	Ta4	TF1	tn12	TN13	TN19	TN23	TN35
1	2	1	1	1	2	1	1	1	1	2	1	1	1	1	1
TN40	Tn9	TQ	TQ12	TQ3	tr10	TR18	TS10	TS17	TS19	TS24	TS25	Ts5	tw10	TW11	TW13
1	1	1	1	1	1	1	1	2	1	1	1	1	1	1	1
TW20	TW4	Tw5	TW5	W3	WA11	WA5	WA8	WD19	WF	WF15	wf2	WF2	WF7	WN4	WN6
1	1	1	1	1	1	1	1	1	1	1	1	2	1	1	1
WN7	WR14	wr2	WR5	WR9	WS1	WS12	Ws5	WS6	WV1	WV2	WV6	yo1	Yo1	Y01	Y010
1	3	1	2	1	1	1	1	1	1	1	2	1	1	2	3
Y011	Y014	Yo15	Y019	Yo21	Y026	Y030	Yo31	Y032							
1	1	1	1	1	1	1	1	1							

The US codes look fine, but the UK should be capitalised.

```
intake_participants <- intake_participants |>
  mutate(
    geo_area_original = geo_area,
    geo_area = str_to_upper(geo_area)
  )
```

That reduced nearly 40 of the possible values.

Age

```
table(intake_participants$age, useNA = "ifany")
```

```
18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37
42 55 94 99 132 125 168 152 168 159 134 135 159 41 46 46 47 42 36 27
38 39 40
22 30 19
```

Nothing to do with age.

Gender

```
table(intake_participants$gender, useNA = "ifany")
```

agender	Agender	Bigender	Trans	Male
1	1			1
Demi-boy	Demiman		gender-fluid	
1	1			1
Genderfluid	He/They			Man
1	1			1280
Non-binary	Prefer not to say		Questioning	
96	3			1
trans woman	Trans woman		Transgender	Man
1	1			1
Transgender Man (FTM)	transmasc lesbian			Woman
1	1			585

I will convert the values to fewer scales for easier analysis (not Man/Woman/Non-binary/Questioning/Other). The number of transgender and individual non-binary identities are too few to be put as their own response groups, as this may cause bias in future modelling. Therefore, I will group gender into the following: Male/Female/Non-binary/Other. Prefer not to say shall be converted to NA, especially as there are only 3.

```
intake_participants <- intake_participants |>
  mutate(
    gender_original = gender,
    gender = str_to_lower(gender_original),
    gender = case_when(
```

```

    grepl("woman", gender, fixed = TRUE) ~ "Woman",
    grepl("agender", gender, fixed = TRUE) |
    grepl("they|them", gender) |
    grepl("demi", gender, fixed = TRUE) |
    grepl("bi", gender, fixed = TRUE) |
    grepl("gender.*fluid", gender) |
    grepl("non.*binary", gender) ~ "Non-binary",
    grepl("man", gender, fixed = TRUE) ~ "Man",
    gender == "prefer not to say" ~ "Prefer not to say",
    TRUE ~ NA_character_
  )
)

table(intake_participants$gender, useNA = "ifany")

```

Man	Non-binary	Prefer not to say	Woman
1282	105	3	587
<NA>			
1			

```

intake_participants |> select(gender, gender_original) |>
  count(gender, gender_original) |> print()

```

```

# A tibble: 18 x 3
  gender      gender_original      n
  <chr>      <chr>          <int>
1 Man       Man             1280
2 Man       Transgender Man      1
3 Man       Transgender Man (FTM) 1
4 Non-binary Agender           1
5 Non-binary Bigender Trans Male  1
6 Non-binary Demi-boy           1
7 Non-binary Demiman             1
8 Non-binary Genderfluid         1
9 Non-binary He/They             1
10 Non-binary Non-binary           96
11 Non-binary agender             1
12 Non-binary gender-fluid         1
13 Non-binary trans masc lesbian  1
14 Prefer not to say Prefer not to say 3

```

15	Woman	Trans woman	1
16	Woman	Woman	585
17	Woman	trans woman	1
18	<NA>	Questioning	1

Education Level

I will make this data ordinal, as then this could be used for future work. This will also involve grouping some of the variables together (Highschool diploma/some college)

```
table(intake_participants$edu_level, useNA = "ifany")
```

	Associate's degree
	Bachelor's degree
	Completed Professional degree
	Completed Secondary education
	Graduate or professional degree (MA, MS, MBA, etc.)
	High school graduate or GED
Higher National Certificate, Higher National Diploma, Bachelor's degree, or post-graduate qualifications	
	Less than a high school diploma or GED
	Master's degree, professional degree, or doctorate
	Preferred
	Some college but no degree
	Some University but no degree
	University Bachelor's degree
	Vocational or technical training

Values according to the codebook: Less than a high school diploma or equivalent; High school graduate or equivalent; Some college but no degree; Associate degree; Bachelor's degree; Master's degree, professional degree, or doctoral degree; Prefer not to say

Also note that “prefer not to say” is being encoded as NA.

```
old_edu <- intake_participants$edu_level
intake_participants$edu_level_original <- old_edu

intake_participants <- intake_participants |>
  mutate(
    edu_level = case_when(
      edu_level_original %in% c(
        "Less than a high school diploma or equivalent",
        "Completed Primary School"
      ) ~ "Below High School",
      edu_level_original %in% c(
        "High school graduate or equivalent",
        "Completed Secondary School"
      ) ~ "High School",
      edu_level_original == "Vocational or Similar" ~ "Vocational or Similar",
      edu_level_original %in% c(
        "Some University but no degree",
        "Some college but no degree"
      ) ~ "Higher education but no degree",
      edu_level_original == "Associate degree" ~ "Associate degree",
      edu_level_original %in% c(
        "Bachelor's degree",
        "University Bachelors Degree",
        "Higher National Certificate, Higher National Diploma, Bachelor's degree, or post-gr"
      ) ~ "Bachelors degree",
      edu_level_original %in% c(
        "Master's degree, professional degree, or doctoral degree",
        "Graduate or professional degree (MA, MS, MBA, PhD, etc)"
      ) ~ "Postgraduate",
      edu_level_original == "Prefer not to say" ~ "Prefer not to say",
      TRUE ~ NA_character_
    )
  )

print(intake_participants |> count(edu_level), n = Inf)
```

```
# A tibble: 8 x 2
```

edu_level	n
<chr>	<int>
1 Associate degree	171
2 Bachelors degree	607
3 Below High School	33
4 High School	432
5 Higher education but no degree	525
6 Postgraduate	197
7 Prefer not to say	6
8 Vocational or Similar	7

Ethnicity

```
table(intake_participants$ethnicity, useNA = "ifany")
```

```

American Indian and Alaska Native alone
      8
Asian alone
    107
Asian or Asian British
    52
Black or African American alone
    130
Black, African, Caribbean or Black British
    15
Mixed or multiple ethnic groups
    28
Native Hawaiian and Other Pacific Islander alone
     2
Other ethnic group
     5
Prefer not to say
     2
Some Other Race alone
    48
Two or More Races
    171
White
    596
White alone

```

770
<NA>
44

I will create the following groups: White, Black, Asian, Native Hawaiian, Mixed, Other, American Indian, Prefer not to say (as NA):

```
ethnicity_old <- intake_participants$ethnicity
intake_participants$ethnicity_original <- ethnicity_old

intake_participants <- intake_participants |>
  mutate(
    ethnicity = ethnicity_original,
    ethnicity = case_when(
      ethnicity %in% c("White alone", "White") ~ "White",
      ethnicity %in% c(
        "Black",
        "Black, African, Caribbean or Black British",
        "Black or African American alone"
      ) ~ "Black",
      ethnicity %in% c(
        "Asian",
        "Asian alone",
        "Asian or Asian British"
      ) ~ "Asian",
      ethnicity %in% c(
        "Mixed",
        "Mixed or multiple ethnic groups",
        "Two or More Races"
      ) ~ "Multiple",
      ethnicity == "American Indian and Alaska Native alone" ~ "American Indian and Alaska Native",
      ethnicity == "Native Hawaiian and Other Pacific Islander alone" ~ "Native Hawaiian and Other Pacific Islander",
      ethnicity == "Prefer not to say" ~ "Prefer not to say",
      is.na(ethnicity) ~ NA_character_,
      TRUE ~ "Other"
    )
  )

table(intake_participants$ethnicity, useNA = "ifany")
```

American Indian and Alaska Native

	8
Asian	159
Black	145
Multiple	199
Native Hawaiian and Other Pacific Islander	2
Other	53
Prefer not to say	2
White	1366
<NA>	44

```
intake_eth_na <- intake_participants |>
  filter(is.na(ethnicity))

intake_participants |>
  mutate(ethnicity_na = is.na(ethnicity)) |>
  select(panel_source, ethnicity_na) |>
  table()
```

	ethnicity_na	
panel_source	FALSE	TRUE
Prolific	1639	44
PureProfile	295	0

```
eth_na_dates <- intake_participants |> filter(is.na(ethnicity)) |>
  mutate(date = as.Date(date)) |>
  pull(date)
```

The rest of the ethnicity missingness is when using prolific data, specifically all on the 26th September 2024, except for one which appears to be refusal. The participants (excluding the one anomaly) were pilot participants and also all part of Prolific. They were evenly distributed across the countries, but all also linked steam. This seems like some form of error in responses, indicating MNAR.

These shall be recoded as simply “Not reported”.

```

intake_participants <- intake_participants |>
  mutate(
    ethnicity = if_else(is.na(ethnicity), "Not reported", ethnicity)
  )

table(intake_participants$ethnicity, useNA = "ifany")

```

American Indian and Alaska Native	8
Asian	159
Black	145
Multiple	199
Native Hawaiian and Other Pacific Islander	2
Not reported	44
Other	53
Prefer not to say	2
White	1366

Martial Status

```

table(intake_participants$marital_status, useNA = "ifany")

```

Co-habiting	1
Cohabiting	3
Dating	1
Divorced	25

engaged	4
Engaged	10
Engaged, to be Married	1
I'm in a committed relationship	1
In a domestic partnership or civil union	243
In a long term relationship	1
in a relationship	2
In a relationship	6
In a Relationship	1
in a relationship but not married	1
In a relationship not married	1
In a relationship, not living together	1
In long-term relationship	1
in relationship, never married	1
Living as married	1
Living with partner	1
living with significant other	1
Long Distance Relationship	1
Long term relationship	1
Long Term Relationship with a British Citizen, living separately	1
Married	266
never married but not single	

	1
Prefer not to say	13
Relationship, live with (non-legal partnership)	1
Separated	11
Separated, intent to divorce, polyamorous relationship with another partner	1
Single, never married	1375

Everything, except one instance, of “Other” is actually just single, by definition of marital status.

```
old_marital_status <- intake_participants$marital_status

intake_participants$marital_status_original <- old_marital_status

intake_participants <- intake_participants |>
  mutate(
    marital_status = marital_status_original,
    marital_status = case_when(
      marital_status == "Married" ~ "Married",
      marital_status == "In a domestic partnership or civil union" ~ "In a domestic partnership or civil union",
      grepl("separated", str_to_lower(marital_status), fixed = TRUE) ~ "Separated",
      marital_status == "Divorced" ~ "Divorced",
      marital_status == "Widowed" ~ "Widowed",
      marital_status == "Prefer not to say" ~ "Prefer not to say",
      is.na(marital_status) ~ NA_character_,
      TRUE ~ "Single, never married",
    )
  )

table(intake_participants$marital_status, useNA = "ifany")
```

Divorced	25
In a domestic partnership or civil union	243
Married	

	266
Prefer not to say	13
Separated	12
Single, never married	1419

Political Party

```
table(intake_participants$political_party, useNA = "ifany")
```

alliance	1
Anarcho communist	1
Apolitical	3
Center	1
Communist	1
Conservative and Unionist Party	30
Democratic Party	591
Democratic Unionist Party (DUP)	1
Either Democratic or Independent	1
Floating voter	1
Green or Socialist	1
Green Party of England and Wales	110
Green Party of Scotland	1
i dont care	1

I dont pay enough attention to politics to know
 1
 I'd pick the egalitarian one, but there isn't one.
 1
 Independent
 428
 Labour Party / Co-operative Party
 220
 Left leaning centrist
 1
 leftist
 3
 Leftist
 1
 Leftist/Green/Progressive
 1
 Liberal Democrats
 68
 libertarian
 1
 Libertarian
 1
 Libertarians
 1
 Marxist-Leninist
 1
 moderate
 1
 No affiliation / Independent
 159
 no political affiliation
 1
 non political
 1
 none
 2
 None
 1
 Not interested in politics
 1
 Peace and Freedom/Socialist
 1
 Plaid Cymru

	7
Prefer not to say	51
Reform UK	55
Republican Party	178
Scottish Green Party	4
scottish greens	1
Scottish Greens	3
Scottish National Party (SNP)	31
Sinn Féin	3
socialist	1
Socialist	2
Socialist party	1
Tactical voting against right wing	1
Unaffiliated	1

Future note to self: I could try split the countries into smaller states/regions (wales/scotland). For now, I will just use the specified parties, not specifically handling the others (socialist, scottish greens).

```
old_political_party <- intake_participants$political_party

intake_participants$political_party_original <- intake_participants$political_party

intake_participants <- intake_participants |>
  mutate(
    political_party = political_party_original,
    political_party = case_when(
      # US
      political_party == "Democratic Party" ~ "Democratic Party",
```

```

political_party == "Republican Party" ~ "Republican Party",
political_party == "Independent" ~ "Independent",

# UK
political_party == "Labour Party / Co-operative Party" ~ "Labour Party / Co-operative Party",
political_party == "Conservative and Unionist Party" ~ "Conservative and Unionist Party",
political_party == "Liberal Democrats" ~ "Liberal Democrats",
political_party == "Scottish National Party (SNP)" ~ "Scottish National Party (SNP)",

political_party == "Sinn Féin" ~ "Sinn Féin",
political_party == "Reform UK" ~ "Reform UK",
political_party == "Democratic Unionist Party (DUP)" ~ "Democratic Unionist Party (DUP)",
political_party == "Green Party of England and Wales" ~ "Green Party of England and Wales",
political_party == "Plaid Cymru" ~ "Plaid Cymru",
political_party == "No affiliation / Independent" ~ "No affiliation / Independent",
str_detect(str_to_lower(political_party), paste(
  "none", "unaffiliated", "no political affiliation", "not interested",
  "apolitical", "non political", "i dont care", "non political",
  "i dont pay enough attention to politics to know",
  "not interested in politics",
  sep = "|"
)) ~ "No affiliation / Independent",

political_party == "Prefer not to say" ~ "Prefer not to say",
is.na(political_party) ~ NA_character_,

# "Anarcho communist", "scottish greens", "All bad"
TRUE ~ "Other"
)
)

print(intake_participants |> count(political_party), n = Inf)

```

```

# A tibble: 15 x 2
  political_party      n
  <chr>            <int>
1 Conservative and Unionist Party    30
2 Democratic Party                591
3 Democratic Unionist Party (DUP)      1
4 Green Party of England and Wales   110
5 Independent                     428
6 Labour Party / Co-operative Party  220

```


7 Liberal Democrats	68
8 No affiliation / Independent	171
9 Other	34
10 Plaid Cymru	7
11 Prefer not to say	51
12 Reform UK	55
13 Republican Party	178
14 Scottish National Party (SNP)	31
15 Sinn Féin	3

Employment

```
table(intake_participants$employment, useNA = "ifany")
```

at-home parent

employed

Full time employment but on the sick

time unpaid carer

I left my job on a voluntary severance program in November 28. I am currently looking for work

I'm currently off work due to a

time studying, and part-time volunteering

(

Pa

S

Employed (variable hours)/disabled

employed

employed, freelancer

Employed/Contractor

time

```
time
```

```
time
```

```
old_employment <- intake_participants$employment

intake_participants$employment_original <- old_employment

intake_participants <- intake_participants |>
  mutate(
    employment = employment_original,
    employment = str_to_lower(employment),
    employment = case_when(
      employment == "working full-time" ~ "Working full-time",
      employment == "working part-time" ~ "Working part-time",
      employment == "not currently employed" ~ "Not currently employed",
      employment == "a homemaker or stay-at-home parent" ~ "A homemaker or stay-at-home parent",
      employment == "student" ~ "Student",
      employment == "retired" ~ "Retired",
      employment == "prefer not to say" ~ "Prefer not to say",
      is.na(employment) ~ NA_character_,
      TRUE ~ "Other"
    )
  )

print(intake_participants |> count(employment), n = Inf)
```

```
# A tibble: 9 x 2
```

employment	n
<chr>	<int>

1 A homemaker or stay-at-home parent	55
2 Not currently employed	373
3 Other	72
4 Prefer not to say	1
5 Retired	2
6 Student	325
7 Working full-time	865
8 Working part-time	284
9 <NA>	1

The sole NA person self reported 600 minutes (10 hours of games) a week. I just want to investigate further:

```
na_emp_pid <- intake_participants |> filter(is.na(employment)) |> pull(pid)

na_emp_steam <- steam |> filter(pid == na_emp_pid)

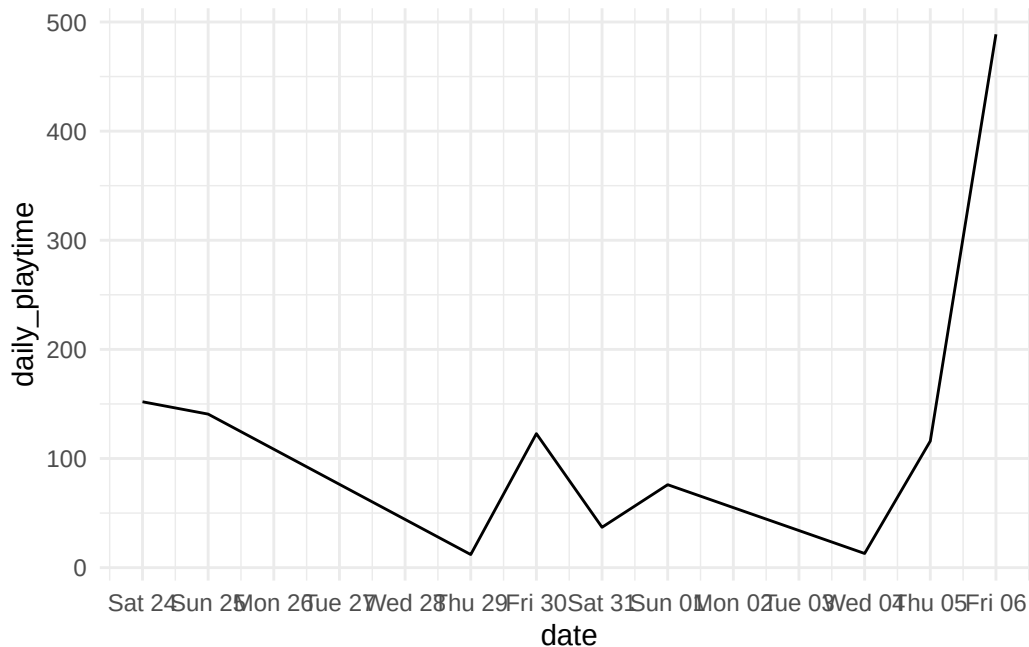
start_survey_date <- min(na_emp_steam$date)
end_period <- start_survey_date + weeks(2)

start_2week_playtime <- na_emp_steam |>
  filter(date >= start_survey_date & date < end_period)

print(paste("Total playtime between ", start_survey_date, " and ", end_period, ": ", sum(sta
```

```
[1] "Total playtime between 2025-05-24 and 2025-06-07 : 1158.163459702"
```

```
start_2week_playtime |>
  group_by(date) |>
  summarise(daily_playtime = sum(minutes)) |>
  ggplot(aes(x = date, y = daily_playtime)) +
  geom_line() +
  scale_x_date(date_labels = "%a %d", breaks = scales::pretty_breaks(n = 14))
```



Check people's ethnicity from similar locations:

```
na_emp <- intake_participants |> filter(is.na(employment))

intake_participants |>
  filter(country == na_emp$country, geo_area == na_emp$geo_area)
```

```
# A tibble: 2 x 95
  pid      date      qualified survey_duration country geo_area
<chr>    <dtm>      <lgl>          <dbl> <chr>    <chr>
1 p5694085417 2025-02-28 06:10:43 TRUE      489 US      633
2 p4503908702 2025-05-22 03:25:06 TRUE      475 US      633
# i 89 more variables: panel_source <chr>, cohort <chr>,
#   diaries_completed <dbl>, panels_completed <dbl>, local_timezone <chr>,
#   age <dbl>, gender <chr>, height <dbl>, weight <dbl>, ethnicity <chr>,
#   edu_level <chr>, employment <chr>, marital_status <chr>,
#   political_party <chr>, self_reported_weekly_play <dbl>,
#   used_platforms <chr>, eligible_platforms <chr>, linked_platforms <chr>,
#   plays_xbox <lgl>, plays_steam <lgl>, plays_nintendo <lgl>, ...
```

I will therefore treat the response of “NA” to imply they are not currently employed:

```
intake_participants <- intake_participants |>
  mutate(
    employment = if_else(is.na(employment), "Not currently employed", employment)
  )

table(intake_participants$employment, useNA = "ifany")
```

A homemaker or stay-at-home parent	Not currently employed
55	374
Other	Prefer not to say
72	1
Retired	Student
2	325
Working full-time	Working part-time
865	284

Missing Data

The rest of the demographic missingness appears to be individuals not disclosing their information, but I will check these instances for any unusual records to be sure:

```
intake_participants <- intake_participants |>
  mutate(
    adhd = case_when(
      !is.na(neuro_diag_adhd) ~ "Diagnosed ADHD",
      is.na(neuro_diag_adhd) & !is.na(neuro_iden_adhd) ~ "Self-Identified ADHD",
      TRUE ~ "Neurotypical"
    )
  )
```

Dependents

```
vis_miss(intake_participants |> select(demographic_vars, starts_with("care_"))) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 0, vjust = 0),
    margins = margin(r = 50, b = 5)
  )
```

Warning: Using an external vector in selections was deprecated in tidysselect 1.1.0.
 i Please use `all\_of()` or `any\_of()` instead.

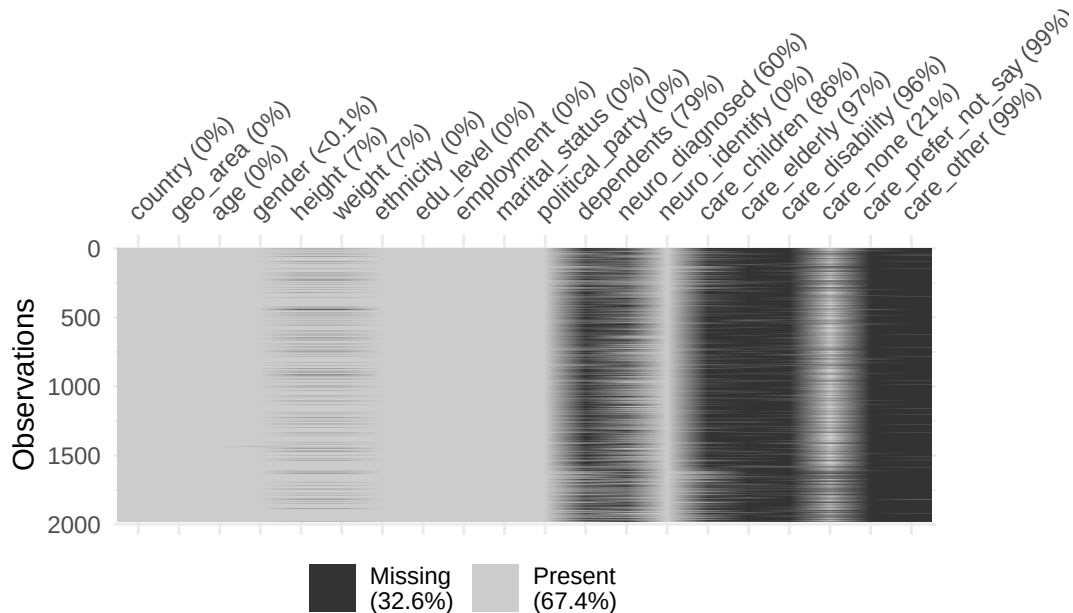
Was:

```
data %>% select(demographic_vars)
```

Now:

```
data %>% select(all_of(demographic_vars))
```

See <<https://tidysselect.r-lib.org/reference/faq-external-vector.html>>.



Interestingly, the proportion of missing dependents is inverse to the proportion of people missing care_none (they don't care for anyone). This makes sense as people who do not care for anyone, will leave dependents blank. Dependents is based on the care variables:

```
addmargins(xtabs(~ dependents + care_none, data = intake_participants, addNA = TRUE))
```

	care_none
dependents	No, I do not have additional caregiving responsibilities
1	0
2	0
3	0
4 or more	0
None	0
Prefer not to say	0

<NA>		1553
Sum		1553
	care_none	
dependents	<NA>	Sum
1	207	207
2	94	94
3	31	31
4 or more	13	13
None	59	59
Prefer not to say	4	4
<NA>	17	1570
Sum	425	1978

We can see that everyone who didn't answer dependents, indicated they did not have additional caregiving responsibilities. This is construct-missingness slightly, and is MAR, as the missingness is heavily dependent on factors such as marital status and age.

I will recode NA dependent participants' "None" if they were answered the care_none variable. For full transparency in further analysis, I will also add a `imputed` flag to the dataset.

```
intake_participants$dependents_original <- intake_participants$dependents
intake_participants <- intake_participants |>
  mutate(
    dependents_original = dependents,
    dependents_imputed = if_else(
      is.na(dependents) & !is.na(care_none),
      1, 0
    ),
    dependents = case_when(
      dependents_imputed == 1 ~ "None",
      TRUE ~ dependents
    )
  )
```

```
addmargins(xtabs(~ dependents + care_children, data = intake_participants, addNA = TRUE))
```

	care_children		
dependents	Yes, I care for children	<NA>	Sum
1	136	71	207
2	87	7	94
3	29	2	31
4 or more	12	1	13

None	6	1606	1612
Prefer not to say	1	3	4
<NA>	0	17	17
Sum	271	1707	1978

```
addmargins(xtabs(~ dependents + care_elderly, data = intake_participants, addNA = TRUE))
```

	care_elderly		
dependents	Yes, I care for an elderly family member	<NA>	Sum
1	31	176	207
2	8	86	94
3	6	25	31
4 or more	1	12	13
None	21	1591	1612
Prefer not to say	1	3	4
<NA>	0	17	17
Sum	68	1910	1978

```
addmargins(xtabs(~ dependents + care_disability, data = intake_participants, addNA = TRUE))
```

	care_disability	
dependents	Yes, I care for a family member with a disability or chronic illness	
1		42
2		6
3		2
4 or more		1
None		19
Prefer not to say		1
<NA>		0
Sum		71

	care_disability	
dependents	<NA>	Sum
1	165	207
2	88	94
3	29	31
4 or more	12	13
None	1593	1612
Prefer not to say	3	4
<NA>	17	17
Sum	1907	1978

```
addmargins(xtabs(~ dependents + care_prefer_not_say, data = intake_participants, addNA = TRUE))
```

	care_prefer_not_say		
dependents	Prefer not to say	<NA>	Sum
1	0	207	207
2	0	94	94
3	0	31	31
4 or more	0	13	13
None	1	1611	1612
Prefer not to say	0	4	4
<NA>	17	0	17
Sum	18	1960	1978

```
addmargins(xtabs(~ dependents + care_other, data = intake_participants, addNA = TRUE))
```

	care_other	
dependents	Cats	I am responsible for the cleaning in my house
1	0	0
2	0	0
3	0	0
4 or more	0	0
None	1	1
Prefer not to say	0	0
<NA>	0	0
Sum	1	1

	care_other	
dependents	I have 6 pets	I have a child on the way
1	0	0
2	0	0
3	0	0
4 or more	0	0
None	1	1
Prefer not to say	0	0
<NA>	0	0
Sum	1	1

	care_other	
dependents	I help care for my family, which includes a child and several animals.	
1		0
2		0
3		0
4 or more		0

None		1
Prefer not to say		0
<NA>		0
Sum		1
	care_other	
dependents	I help care for the pets in my household	
1		0
2		0
3		0
4 or more		0
None		1
Prefer not to say		0
<NA>		0
Sum		1
	care_other	
dependents	I help my elderly parents shop when they come into town since they live o	
1		
2		
3		
4 or more		
None		
Prefer not to say		
<NA>		
Sum		
	care_other	
dependents	I may be moving home within a couple of months to take on caregiver resp	
1		
2		
3		
4 or more		
None		
Prefer not to say		
<NA>		
Sum		
	care_other	
dependents	I occasionally have caregiving responsibilities for a friend	
1		0
2		0
3		0
4 or more		0
None		1
Prefer not to say		0
<NA>		0

Sum		1
-----	--	---

	care_other	
dependents	I take care of 3 ferrets on a daily basis, if that counts.	
1		0
2		0
3		0
4 or more		0
None		1
Prefer not to say		0
<NA>		0
Sum		1

	care_other	
dependents	I take care of my husband	
1		0
2		0
3		0
4 or more		0
None		1
Prefer not to say		0
<NA>		0
Sum		1

	care_other	
dependents	I work for a facility and do minor caregiving	
1		0
2		0
3		0
4 or more		0
None		1
Prefer not to say		0
<NA>		0
Sum		1

	care_other	
dependents	I'm the chronically ill one	
1		0
2		0
3		0
4 or more		0
None		1
Prefer not to say		0
<NA>		0
Sum		1

	care_other	
dependents	Not full-time but I informally take care of two elderly family members.	

1				1
2				0
3				0
4 or more				0
None				0
Prefer not to say				0
<NA>				0
Sum				1
	care_other			
dependents	Not yet, but in the near future, care for children	pet	dog	
1		0	0	
2		0	0	
3		0	0	
4 or more		0	0	
None		1	1	
Prefer not to say		0	0	
<NA>		0	0	
Sum		1	1	
	care_other			
dependents	Take care of 14 year old sibling			
1		1		
2		0		
3		0		
4 or more		0		
None		0		
Prefer not to say		0		
<NA>		0		
Sum		1		
	care_other			
dependents	They're not diagnosed with anything, but I do a lot of caregiving for a			
1				
2				
3				
4 or more				
None				
Prefer not to say				
<NA>				
Sum				
	care_other			
dependents	Yes I care for chickens and dogs			
1		0		
2		0		
3		0		

4 or more	1
None	0
Prefer not to say	0
<NA>	0
Sum	1

	care_other
dependents	Yes, I help watch over a family member that may need assistance with dai.
1	
2	
3	
4 or more	
None	
Prefer not to say	
<NA>	
Sum	

	care_other	
dependents	<NA>	Sum
1	205	207
2	94	94
3	31	31
4 or more	12	13
None	1596	1612
Prefer not to say	3	4
<NA>	17	17
Sum	1958	1978

The final missing 17 rows are all when people responded to “prefer not to say” for care-giving abilities. I will leave these as is, rather than dropping them as they don’t even account for 1% of the data.

Height and Weight

Now I will investigate the missing weight and height data:

```
na_weight <- if_else(is.na(intake_participants$weight), 1, 0)
na_height <- if_else(is.na(intake_participants$height), 1, 0)

xtabs(~na_height + na_weight)
```

	na_weight	
na_height	0	1
0 1840	0	

1 1 137

There is only one record where height is missing and weight isn't, 1840 are not missing, 137 records are missing both.

```
check_cols <- demographic_vars[!demographic_vars %in% c("height", "weight", "geo_area")]
for (col in check_cols) {
  cat("Cross-tab for", col, "and na_weight:\n")
  form_string <- paste0("~ ", col, " + na_weight")
  xtable <- xtabs(as.formula(form_string), data = intake_participants, addNA = TRUE)
  na_pct <- round(prop.table(xtable, margin = 1)[, "1"] * 100, 1)
  na_total_pct <- round(xtable[, "1"] / sum(xtable[, "1"]) * 100, 1)
  print(cbind(xtable, pct_na = na_pct, tot_pct_na = na_total_pct))
}
```

Cross-tab for country and na_weight:

	0	1	pct_na	tot_pct_na
UK	655	64	8.9	46.7
US	1186	73	5.8	53.3

Cross-tab for age and na_weight:

	0	1	pct_na	tot_pct_na
18	39	3	7.1	2.2
19	50	5	9.1	3.6
20	86	8	8.5	5.8
21	96	3	3.0	2.2
22	119	13	9.8	9.5
23	117	8	6.4	5.8
24	156	12	7.1	8.8
25	137	15	9.9	10.9
26	156	12	7.1	8.8
27	150	9	5.7	6.6
28	123	11	8.2	8.0
29	126	9	6.7	6.6
30	149	10	6.3	7.3
31	37	4	9.8	2.9
32	44	2	4.3	1.5
33	45	1	2.2	0.7
34	45	2	4.3	1.5
35	40	2	4.8	1.5
36	34	2	5.6	1.5
37	27	0	0.0	0.0
38	19	3	13.6	2.2

39 28 2 6.7 1.5
 40 18 1 5.3 0.7

Cross-tab for gender and na_weight:

	0	1	pct_na	tot_pct_na
Man	1205	77	6.0	56.2
Non-binary	91	14	13.3	10.2
Prefer not to say	0	3	100.0	2.2
Woman	544	43	7.3	31.4
<NA>	1	0	0.0	0.0

Cross-tab for ethnicity and na_weight:

	0	1	pct_na	tot_pct_na
American Indian and Alaska Native	7	1	12.5	0.7
Asian	142	17	10.7	12.4
Black	137	8	5.5	5.8
Multiple	192	7	3.5	5.1
Native Hawaiian and Other Pacific Islander	2	0	0.0	0.0
Not reported	42	2	4.5	1.5
Other	49	4	7.5	2.9
Prefer not to say	0	2	100.0	1.5
White	1270	96	7.0	70.1

Cross-tab for edu_level and na_weight:

	0	1	pct_na	tot_pct_na
Associate degree	160	11	6.4	8.0
Bachelors degree	566	41	6.8	29.9
Below High School	28	5	15.2	3.6
High School	401	31	7.2	22.6
Higher education but no degree	492	33	6.3	24.1
Postgraduate	186	11	5.6	8.0
Prefer not to say	1	5	83.3	3.6
Vocational or Similar	7	0	0.0	0.0

Cross-tab for employment and na_weight:

	0	1	pct_na	tot_pct_na
A homemaker or stay-at-home parent	55	0	0.0	0.0
Not currently employed	338	36	9.6	26.3
Other	61	11	15.3	8.0
Prefer not to say	0	1	100.0	0.7
Retired	1	1	50.0	0.7
Student	305	20	6.2	14.6
Working full-time	816	49	5.7	35.8
Working part-time	265	19	6.7	13.9

Cross-tab for marital_status and na_weight:

	0	1	pct_na	tot_pct_na
Divorced	25	0	0.0	0.0

In a domestic partnership or civil union	229	14	5.8	10.2
Married	254	12	4.5	8.8
Prefer not to say	6	7	53.8	5.1
Separated	12	0	0.0	0.0
Single, never married	1315	104	7.3	75.9

Cross-tab for political_party and na_weight:

	0	1	pct_na	tot_pct_na
Conservative and Unionist Party	28	2	6.7	1.5
Democratic Party	557	34	5.8	24.8
Democratic Unionist Party (DUP)	1	0	0.0	0.0
Green Party of England and Wales	97	13	11.8	9.5
Independent	405	23	5.4	16.8
Labour Party / Co-operative Party	201	19	8.6	13.9
Liberal Democrats	65	3	4.4	2.2
No affiliation / Independent	153	18	10.5	13.1
Other	32	2	5.9	1.5
Plaid Cymru	6	1	14.3	0.7
Prefer not to say	39	12	23.5	8.8
Reform UK	52	3	5.5	2.2
Republican Party	171	7	3.9	5.1
Scottish National Party (SNP)	31	0	0.0	0.0
Sinn Féin	3	0	0.0	0.0

Cross-tab for dependents and na_weight:

	0	1	pct_na	tot_pct_na
1	202	5	2.4	3.6
2	92	2	2.1	1.5
3	30	1	3.2	0.7
4 or more	13	0	0.0	0.0
None	1491	121	7.5	88.3
Prefer not to say	3	1	25.0	0.7
<NA>	10	7	41.2	5.1

Cross-tab for neuro_diagnosed and na_weight:

	0	1	pct_na	tot_pct_na
No	239	22	8.4	16.1
Prefer not to say	21	8	27.6	5.8
Yes	471	30	6.0	21.9
<NA>	1110	77	6.5	56.2

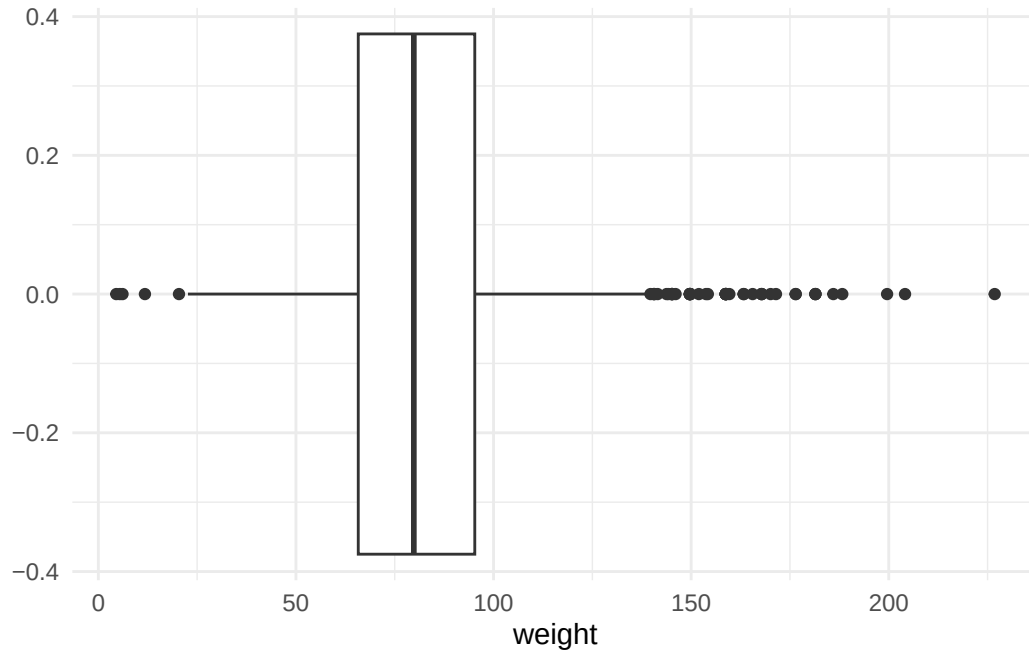
Cross-tab for neuro_identify and na_weight:

	0	1	pct_na	tot_pct_na
No	1056	65	5.8	47.4
Prefer not to say	54	12	18.2	8.8
Yes	731	60	7.6	43.8

Now let's just get a general summary of height and weight:

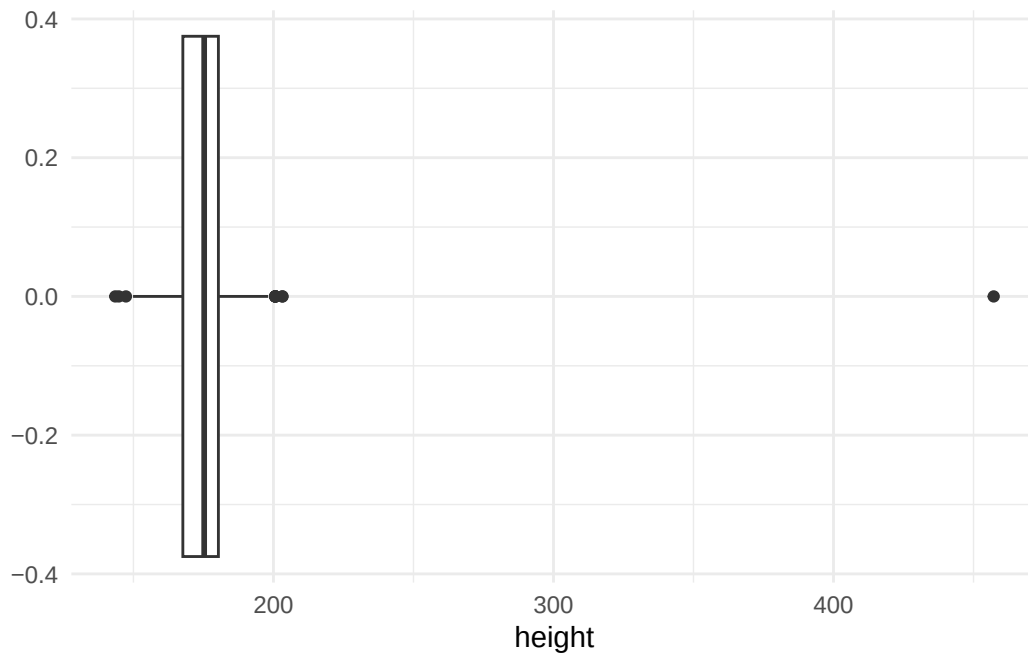
```
intake_participants |> select(height, weight) |>  
  ggplot(aes(x = weight)) +  
  geom_boxplot()
```

Warning: Removed 137 rows containing non-finite outside the scale range
(`stat\_boxplot()`).



```
intake_participants |> select(height, weight) |>  
  ggplot(aes(x = height)) +  
  geom_boxplot()
```

Warning: Removed 138 rows containing non-finite outside the scale range
(`stat\_boxplot()`).



Let me look at all the outlying values from these columns:

```
# IQR outlier method
height_quartiles <- quantile(intake_participants$height, probs = c(.25, .5, .75), na.rm = TRUE)
weight_quartiles <- quantile(intake_participants$weight, probs = c(.25, .5, .75), na.rm = TRUE)

height_iqr <- IQR(intake_participants$height, na.rm = TRUE)
weight_iqr <- IQR(intake_participants$weight, na.rm = TRUE)

outlying_height_weight <- intake_participants |>
  filter(
    height < (height_quartiles[1] - 1.5 * height_iqr) |
    height > (height_quartiles[3] + 1.5 * height_iqr) |
    weight < (weight_quartiles[1] - 1.5 * weight_iqr) |
    weight > (weight_quartiles[3] + 1.5 * weight_iqr)
  ) |>
  select(pid, age, country, height, weight, edu_level, marital_status, employment)
```

I can try impute these, or simply leave them in the binary missingness indicator. The data is missing at an item-level -

```
missing_weight <- intake_participants |> filter(is.na(weight))

missing_weight |>
```

```
summarise(across(everything(), ~sum(is.na(.x)))) |>
pivot_longer(everything(), names_to = "variable", values_to = "na_count") |>
arrange(-na_count) |>
mutate(pct_na = round(na_count / nrow(missing_weight), 2))
```

```
# A tibble: 98 x 3
  variable          na_count pct_na
  <chr>          <int>   <dbl>
1 height             137     1
2 weight             137     1
3 neuro_diag_tourette 137     1
4 neuro_diag_nvld     137     1
5 neuro_diag_prefer_not_say 137     1
6 neuro_iden_tourette 137     1
7 neuro_iden_nvld     137     1
8 neuro_iden_other    137     1
9 life_sat           137     1
10 wemwbs_1           137     1
# i 88 more rows
```

It appears that the weight variable has no real relation to any other variables' missingness. This indicates the data is missing at the item-level. This seems to be bordering between MAR and MNAR - people can be sensitive to disclosing their weight and height.

Does gender affect weight's missingness:

```
# remove the 4 participants with missing gender
clean_data <- subset(intake_participants, !is.na(gender))
clean_data <- clean_data |> filter(gender != "Prefer not to say") # PNTS only has 3 groups
weight_gender_table <- table(clean_data$gender, is.na(clean_data$weight), useNA = "ifany")
chisq.test(weight_gender_table)
```

Pearson's Chi-squared test

```
data: weight_gender_table
X-squared = 8.6154, df = 2, p-value = 0.01346
```

```
weight_gender_table
```

	FALSE	TRUE
Man	1205	77
Non-binary	91	14
Woman	544	43

```
height_gender_table <- table(clean_data$gender, is.na(clean_data$height), useNA = "ifany")
chisq.test(height_gender_table)
```

Pearson's Chi-squared test

```
data: height_gender_table
X-squared = 8.3151, df = 2, p-value = 0.01565
```

```
height_gender_table
```

	FALSE	TRUE
Man	1204	78
Non-binary	91	14
Woman	544	43

This confirms the weight data is MAR, highlighting nearly 16% of Non-binary participants are missing weight.

I will find variables suitable to be used in multiple imputation (MICE):

```
trialling_cols <- c("pid", "gender", "weight", "height", "age", "ethnicity", "country", "edu_
complete_cases_df <- intake_participants %>%
  filter(!is.na(weight) & !is.na(self_reported_weekly_play)) |>
  mutate(across(c(gender, ethnicity, country, edu_level, employment), as.factor)) |>
  select(all_of(trialling_cols))

# Fit a temporary diagnostic linear model
vif_model <- lm(weight ~ gender + age + ethnicity + country + edu_level + employment + self_r
               data = complete_cases_df)

# Check VIF scores
vif(vif_model)
```

	GVIF	Df	GVIF ^{1/(2*Df)}
gender	1.122566	2	1.029326
age	1.317448	1	1.147801
ethnicity	1.242076	7	1.015605
country	1.152531	1	1.073560
edu_level	1.289068	6	1.021385
employment	1.617096	6	1.040866
self_reported_weekly_play	1.111510	1	1.054282

Features lack multi-collinearity.

```
# pearson correlation for numeric
correlation_data <- intake_participants |> select(age, self_reported_weekly_play, weight) |>
  mutate(na_weight = is.na(weight))
cor(correlation_data[, c("age", "self_reported_weekly_play", "na_weight")], use = "complete.")
```

	age	self_reported_weekly_play	na_weight
age	1.00000000	-0.06689052	-0.04860591
self_reported_weekly_play	-0.06689052	1.00000000	-0.02982988
na_weight	-0.04860591	-0.02982988	1.00000000

```
# kruskal-wallis test for categoric
for (col in c("country", "gender", "ethnicity", "employment", "edu_level", "marital_status")) {
  # col_weight_formula <- as.formula(paste0("~weight+", col))
  # print(addmargins(xtabs(col_weight_formula, data = intake_participants)))
  print(kruskal.test(as.formula(paste0("weight~", col)), data = intake_participants))
}
```

Kruskal-Wallis rank sum test

data: weight by country

Kruskal-Wallis chi-squared = 1.5513, df = 1, p-value = 0.2129

Kruskal-Wallis rank sum test

data: weight by gender

Kruskal-Wallis chi-squared = 97.141, df = 2, p-value < 2.2e-16

Kruskal-Wallis rank sum test

data: weight by ethnicity

Kruskal-Wallis chi-squared = 53.423, df = 7, p-value = 3.059e-09

Kruskal-Wallis rank sum test

data: weight by employment

Kruskal-Wallis chi-squared = 35.511, df = 6, p-value = 3.431e-06

Kruskal-Wallis rank sum test

data: weight by edu_level

Kruskal-Wallis chi-squared = 8.4708, df = 7, p-value = 0.2929

Kruskal-Wallis rank sum test

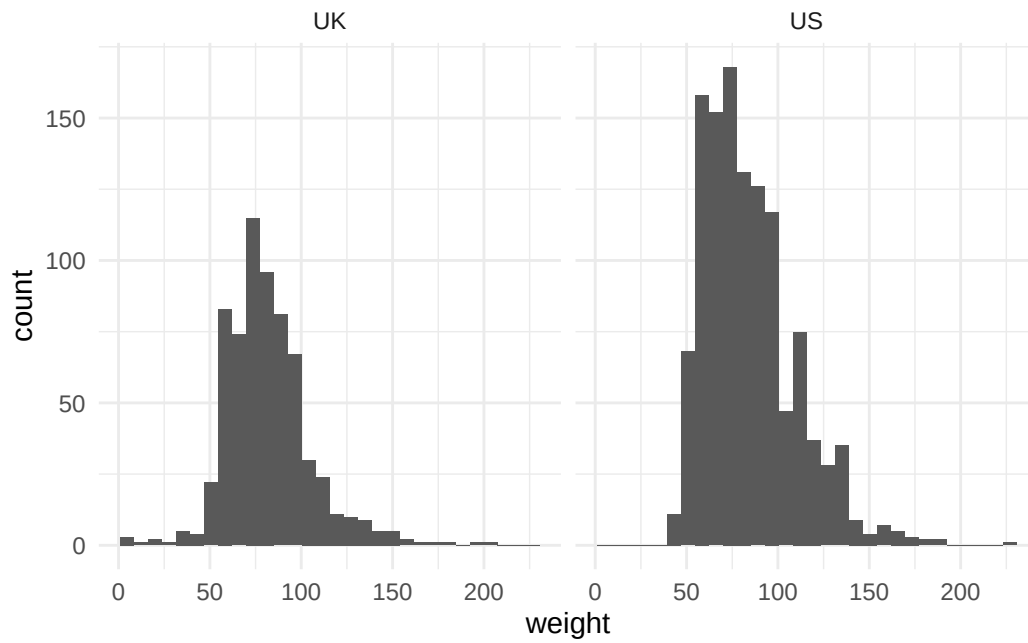
data: weight by marital_status

Kruskal-Wallis chi-squared = 17.632, df = 5, p-value = 0.003445

```
# kruskal-wallis test for categoric
for (col in c("country", "gender", "ethnicity", "employment", "edu_level", "marital_status")) {
  print(ggplot(intake_participants, aes(x = weight)) +
    geom_histogram() +
    facet_wrap(vars(.data[[col]])))
}
```

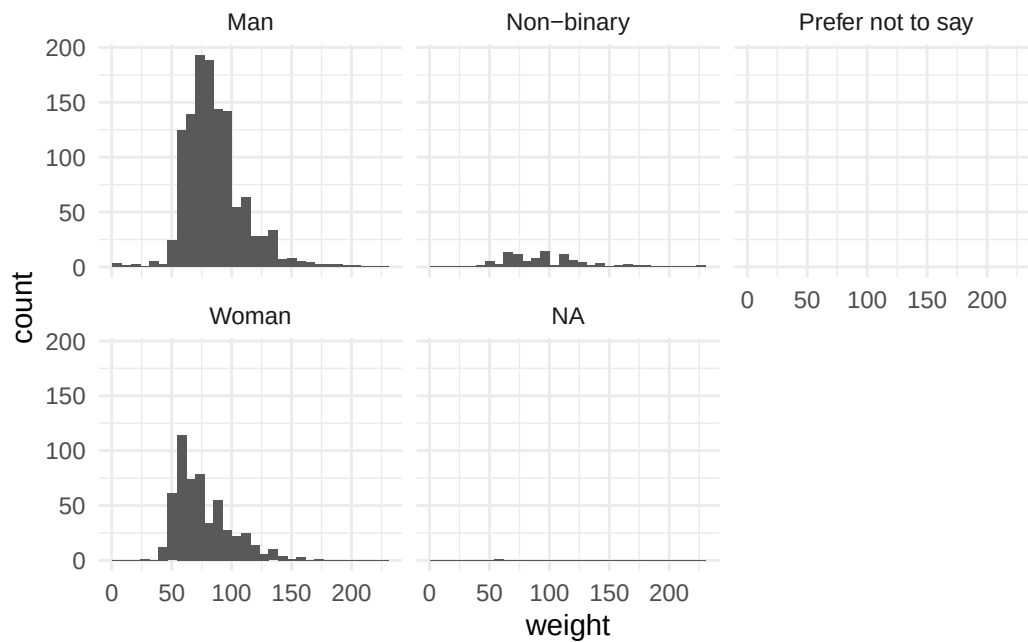
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

Warning: Removed 137 rows containing non-finite outside the scale range (``stat_bin()``).



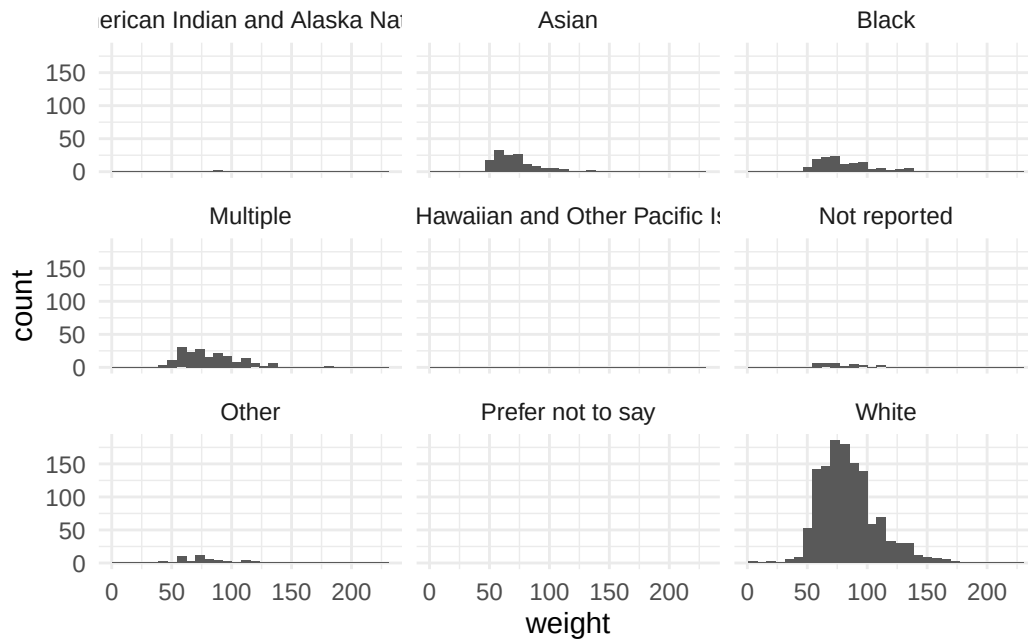
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

Warning: Removed 137 rows containing non-finite outside the scale range (``stat_bin()``).



```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

Warning: Removed 137 rows containing non-finite outside the scale range
(`stat\_bin()`).



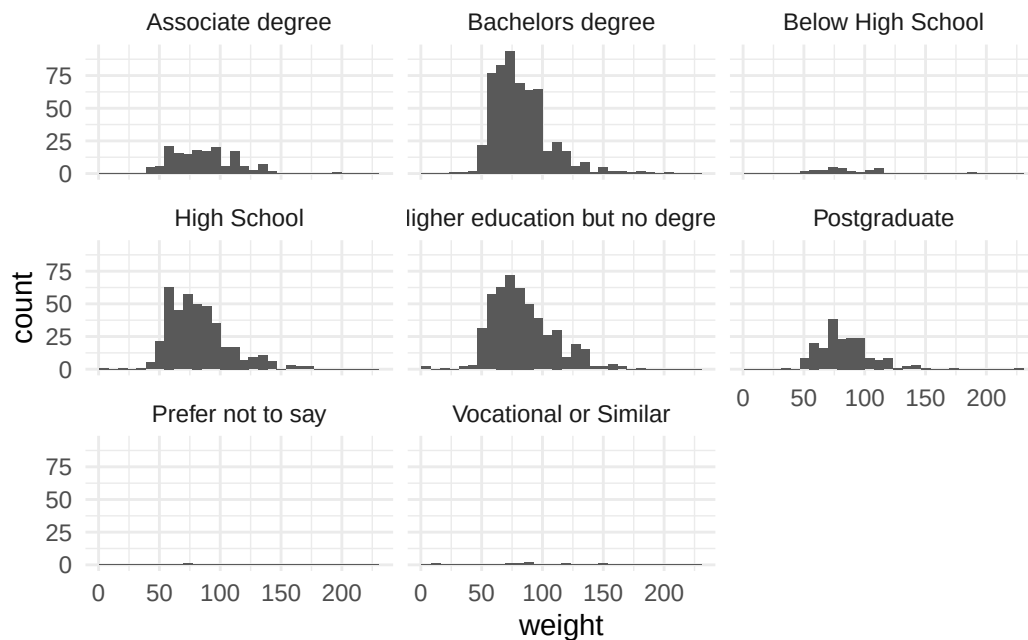
```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

Warning: Removed 137 rows containing non-finite outside the scale range
(`stat\_bin()`).



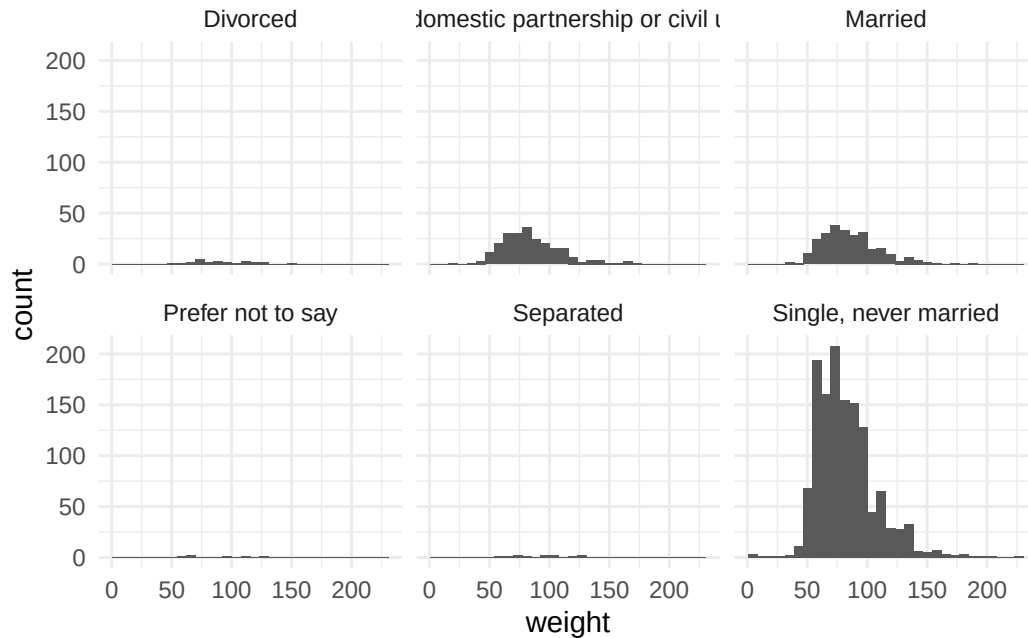
``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

Warning: Removed 137 rows containing non-finite outside the scale range (``stat_bin()``).



``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

Warning: Removed 137 rows containing non-finite outside the scale range (``stat_bin()``).



I will perform multiple imputation with gender, ethnicity, employment, and marital status:

```
# check collinearity
impute_subset <- intake_participants |>
  select(
    pid,
    # targets
    height, weight,
    # Core biological/demographic predictors
    gender, age, ethnicity,
    # Geographical/Socioeconomic predictors
    country, edu_level, employment,
  ) |>
# Convert all textual categorical fields to factors (Required by MICE)
mutate(
  # remove anomalies quickly - they will be imputed
  weight = if_else(weight < 30, NA_real_, weight),
  height = if_else(height > 400, NA_real_, height),
```

```

    across(where(is.character), as.factor)
  )

mice_config <- mice(impute_subset, maxit = 0)

# which column was used for imputing the row feature
pred_matrix <- mice_config$predictorMatrix

# force height and weight to be the only features imputed
no_impute_cols <- c("gender", "age", "ethnicity", "country", "edu_level", "employment")
pred_matrix[no_impute_cols, ] <- 0
pred_matrix["pid",] <- 0
pred_matrix[, "pid"] <- 0

# force non-impute cols to have no method
mice_methods <- mice_config$method
mice_methods[] <- ""
mice_methods[c("height", "weight")] <- "pmm"

# 'pmm' (Predictive Mean Matching) handles missing continuous variables seamlessly
imputed_output <- mice(
  data = impute_subset,
  m = 5, # 5 distinct imputed datasets
  method = mice_methods,
  predictorMatrix = pred_matrix,
  seed = 30
)

```

```

iter imp variable
1 1 height weight
1 2 height weight
1 3 height weight
1 4 height weight
1 5 height weight
2 1 height weight
2 2 height weight
2 3 height weight
2 4 height weight
2 5 height weight
3 1 height weight
3 2 height weight

```

```

3 3 height weight
3 4 height weight
3 5 height weight
4 1 height weight
4 2 height weight
4 3 height weight
4 4 height weight
4 5 height weight
5 1 height weight
5 2 height weight
5 3 height weight
5 4 height weight
5 5 height weight

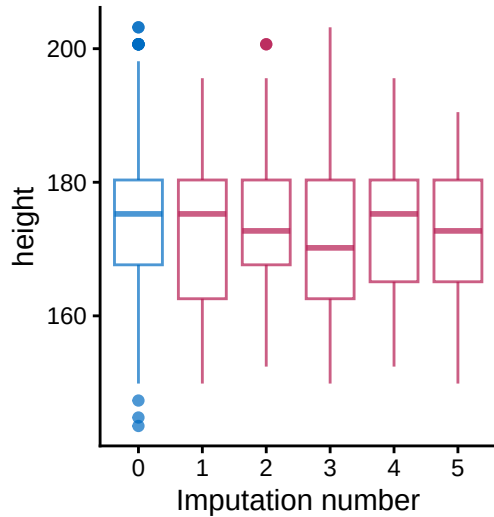
```

Warning: Number of logged events: 50

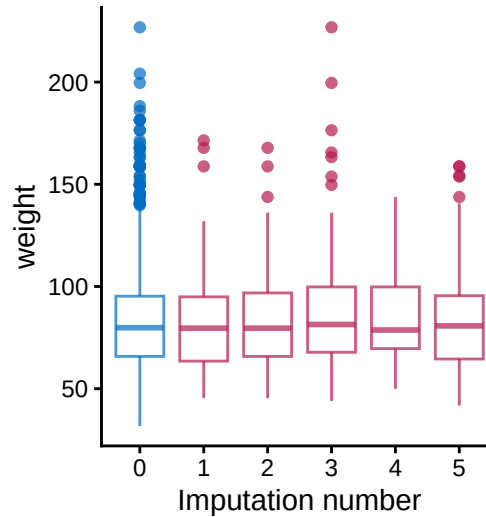
```

map(c("height", "weight"), ~{
  ggmicie(imputed_output, aes(x = .imp, y = .data[.x])) +
    geom_boxplot() +
    labs(x = "Imputation number")
}) |>
  wrap_plots()

```



observed imputed



observed imputed

```
print("Means of original data:")
```

```
[1] "Means of original data:"
```

```
intake_participants |> select(height, weight) |> colMeans(na.rm = TRUE)
```

```
      height      weight  
173.88839  83.44433
```

```
print("Means of imputed data:")
```

```
[1] "Means of imputed data:"
```

```
imputed_output |>  
  mice::complete("all", include = FALSE) |>  
  map(select, height, weight) |>  
  map(colMeans)
```

```
$`1`  
      height      weight  
173.66755  83.59253
```

```
$`2`  
      height      weight  
173.73843  83.78811
```

```
$`3`  
      height      weight  
173.60623  83.92512
```

```
$`4`  
      height      weight  
173.71249  83.75575
```

```
$`5`  
      height      weight  
173.6168  83.7055
```

Now I will merge these with the dataset:

```

imputed_df <- complete(imputed_output, 1)

imputed_cols <- imputed_df |>
  select(pid, imputed_height = height, imputed_weight = weight)

intake_participants$height_original <- intake_participants$height
intake_participants$weight_original <- intake_participants$weight

intake_participants <- intake_participants |>
  left_join(imputed_cols, by = "pid") |>
  mutate(
    height_imputed = if_else(is.na(height), 1, 0),
    weight_imputed = if_else(is.na(weight), 1, 0),
    height = replace_na(imputed_height),
    weight = replace_na(imputed_weight)
  ) |>
  select(-imputed_weight, -imputed_height)

summary(intake_participants)

```

pid	date	qualified
Length:1978	Min. :2024-09-26 19:20:41	Mode:logical
Class :character	1st Qu.:2025-02-27 19:56:08	TRUE:1978
Mode :character	Median :2025-03-01 07:19:11	
	Mean :2025-03-23 15:55:37	
	3rd Qu.:2025-05-29 03:28:03	
	Max. :2025-06-09 09:05:44	

survey_duration	country	geo_area	panel_source
Min. : 46.0	Length:1978	Length:1978	Length:1978
1st Qu.: 393.2	Class :character	Class :character	Class :character
Median : 588.0	Mode :character	Mode :character	Mode :character
Mean : 779.2			
3rd Qu.: 928.8			
Max. :13428.0			

cohort	diaries_completed	panels_completed	local_timezone
Length:1978	Min. : 0.000	Min. :0.000	Length:1978
Class :character	1st Qu.: 0.000	1st Qu.:1.000	Class :character
Mode :character	Median : 0.000	Median :4.000	Mode :character
	Mean : 8.304	Mean :3.312	
	3rd Qu.:21.000	3rd Qu.:6.000	

Max. :30.000 Max. :6.000

age	gender	height	weight
Min. :18.00	Length:1978	Min. :143.5	Min. : 31.75
1st Qu.:23.00	Class :character	1st Qu.:165.1	1st Qu.: 65.77
Median :26.00	Mode :character	Median :175.3	Median : 79.83
Mean :26.72		Mean :173.7	Mean : 83.59
3rd Qu.:30.00		3rd Qu.:180.3	3rd Qu.: 95.25
Max. :40.00		Max. :203.2	Max. :226.80

ethnicity	edu_level	employment	marital_status
Length:1978	Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

political_party	self_reported_weekly_play	used_platforms
Length:1978	Min. : 30	Length:1978
Class :character	1st Qu.: 600	Class :character
Mode :character	Median : 900	Mode :character
	Mean :1083	
	3rd Qu.:1500	
	Max. :3000	
	NA's :499	

eligible_platforms	linked_platforms	plays_xbox	plays_steam
Length:1978	Length:1978	Mode :logical	Mode :logical
Class :character	Class :character	FALSE:1446	FALSE:339
Mode :character	Mode :character	TRUE :532	TRUE :1639

plays_nintendo	plays_ios	plays_android	plays_playstation
Mode :logical	Mode :logical	Mode :logical	Mode :logical
FALSE:1235	FALSE:1432	FALSE:1512	FALSE:1520
TRUE :743	TRUE :546	TRUE :466	TRUE :458

prop_steam	prop_xbox	prop_playstation	prop_nintendo
------------	-----------	------------------	---------------

Min. : 0.00	Min. : 0.00	Min. : 0.00	Min. : 0.00
1st Qu.: 50.00	1st Qu.: 12.00	1st Qu.: 7.00	1st Qu.: 7.00
Median : 80.00	Median : 40.00	Median :18.00	Median : 20.00
Mean : 72.23	Mean : 44.88	Mean :22.66	Mean : 30.21
3rd Qu.:100.00	3rd Qu.: 75.00	3rd Qu.:33.00	3rd Qu.: 41.00
Max. :100.00	Max. :100.00	Max. :91.00	Max. :100.00
NA's :339	NA's :1441	NA's :1525	NA's :1237

prop_ios	prop_android	prop_other	nintendo_uses_family
Min. : 0.00	Min. : 0.00	Min. : 0.00	Length:1978
1st Qu.: 7.00	1st Qu.: 9.00	1st Qu.:19.25	Class :character
Median :14.00	Median :15.00	Median :40.00	Mode :character
Mean :20.01	Mean :20.32	Mean :39.24	
3rd Qu.:28.00	3rd Qu.:28.75	3rd Qu.:60.00	
Max. :95.00	Max. :90.00	Max. :85.00	
NA's :1432	NA's :1512	NA's :1920	

android_uses_family	ios_uses_family	steam_uses_family	xbox_uses_family
Length:1978	Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

total_recent_play_accounted_for	total_play_possible_to_collect
Min. : 0.00	Min. : 0.00
1st Qu.: 0.00	1st Qu.: 54.25
Median : 50.00	Median : 88.00
Mean : 47.64	Mean : 73.79
3rd Qu.: 95.00	3rd Qu.:100.00
Max. :100.00	Max. :100.00

neuro_identify	neuro_diagnosed	neuro_diag_asd	neuro_diag_adhd
Length:1978	Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

neuro_diag_dyslexia	neuro_diag_dyspraxia	neuro_diag_dyscalculia
Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

neuro_diag_tourette	neuro_diag OCD	neuro_diag SPD	neuro_diag NVLD
Length:1978	Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

neuro_diag_specific_ld	neuro_diag_prefer_not_say	neuro_diag_other
Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

neuro_iden ASD	neuro_iden ADHD	neuro_iden dyslexia	neuro_iden dyspraxia
Length:1978	Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

neuro_iden_dyscalculia	neuro_iden_tourette	neuro_iden OCD
Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

neuro_iden SPD	neuro_iden NVLD	neuro_iden_specific_ld
Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

neuro_iden_prefer_not_say	neuro_iden_other	care_children
Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

care_elderly	care_disability	care_none	care_prefer_not_say
Length:1978	Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

care_other	dependents	life_sat	wemwbs_1
Length:1978	Length:1978	Min. : NA	Length:1978
Class :character	Class :character	1st Qu.: NA	Class :character
Mode :character	Mode :character	Median : NA	Mode :character
		Mean :NaN	
		3rd Qu.: NA	
		Max. : NA	
		NA's :1978	
wemwbs_2	wemwbs_3	wemwbs_4	wemwbs_5
Length:1978	Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

wemwbs_6	wemwbs_7	affective_valence
Length:1978	Length:1978	Min. : NA
Class :character	Class :character	1st Qu.: NA
Mode :character	Mode :character	Median : NA
		Mean :NaN
		3rd Qu.: NA
		Max. : NA
		NA's :1978

nintendo_account_creation_date	nintendo_first_party_prop	geo_area_original
Min. :2016-02-17 09:28:57	Min. :0.0000	Length:1978

1st Qu.:	2017-12-26 07:24:09	1st Qu.:	0.3294	Class :	character
Median :	2019-09-04 00:15:09	Median :	0.6382	Mode :	character
Mean :	2019-10-27 13:53:08	Mean :	0.5977		
3rd Qu.:	2020-12-19 15:36:01	3rd Qu.:	0.8837		
Max. :	2025-06-04 09:06:37	Max. :	1.0000		
NA's :	1382	NA's :	1389		
gender_original	edu_level_original	ethnicity_original			
Length:1978	Length:1978	Length:1978			
Class :character	Class :character	Class :character			
Mode :character	Mode :character	Mode :character			

marital_status_original	political_party_original	employment_original
Length:1978	Length:1978	Length:1978
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

adhd	dependents_original	dependents_imputed	height_original
Length:1978	Length:1978	Min. :0.0000	Min. :143.5
Class :character	Class :character	1st Qu.:1.0000	1st Qu.:167.6
Mode :character	Mode :character	Median :1.0000	Median :175.3
		Mean :0.7851	Mean :173.9
		3rd Qu.:1.0000	3rd Qu.:180.3
		Max. :1.0000	Max. :457.2
			NA's :138
weight_original	height_imputed	weight_imputed	
Min. : 4.54	Min. :0.00000	Min. :0.00000	
1st Qu.: 65.77	1st Qu.:0.00000	1st Qu.:0.00000	
Median : 79.83	Median :0.00000	Median :0.00000	
Mean : 83.44	Mean :0.06977	Mean :0.06926	
3rd Qu.: 95.25	3rd Qu.:0.00000	3rd Qu.:0.00000	
Max. :226.80	Max. :1.00000	Max. :1.00000	
NA's :137			

Exploring the demographics

```
# TODO: finish exploring care variables
# intake_participants$care <- intake_participants |> case_when(
#   !is.na(care_children) ~ "Yes",
#   !is.na(care_elderly) ~ "Yes",
#   !is.na(care_disability) ~ "Yes",
#   !is.na(care_other) ~ ""
# )

demographic_vars <- c(
  "country",
  "geo_area",
  "age",
  "gender",
  "height",
  "weight",
  "ethnicity",
  "edu_level",
  "employment",
  "marital_status",
  "political_party",
  "dependents",
  "neuro_diagnosed",
  "neuro_identify"
)
```

Descriptive Statistics

```
intake_participants |>
  select(age, height, weight, gender, ethnicity, country, edu_level) |>
  tbl_summary(
    statistic = list(
      all_continuous() ~ "{mean} ({sd}) [{min}, {max}]",
      all_categorical() ~ "{n} / {N} ({p}%)"
    ),
    missing = "ifany"
  ) |>
  bold_labels() |>
  modify_header(label = "**Variable**")
```

Variable	N = 1,978 ¹
age	26.7 (5.0) [18.0, 40.0]
height	174 (10) [144, 203]
weight	84 (24) [32, 227]
gender	
Man	1,282 / 1,977 (65%)
Non-binary	105 / 1,977 (5.3%)
Prefer not to say	3 / 1,977 (0.2%)
Woman	587 / 1,977 (30%)
Unknown	1
ethnicity	
American Indian and Alaska Native	8 / 1,978 (0.4%)
Asian	159 / 1,978 (8.0%)
Black	145 / 1,978 (7.3%)
Multiple	199 / 1,978 (10%)
Native Hawaiian and Other Pacific Islander	2 / 1,978 (0.1%)
Not reported	44 / 1,978 (2.2%)
Other	53 / 1,978 (2.7%)
Prefer not to say	2 / 1,978 (0.1%)
White	1,366 / 1,978 (69%)
country	
UK	719 / 1,978 (36%)
US	1,259 / 1,978 (64%)
edu_level	
Associate degree	171 / 1,978 (8.6%)
Bachelors degree	607 / 1,978 (31%)
Below High School	33 / 1,978 (1.7%)
High School	432 / 1,978 (22%)
Higher education but no degree	525 / 1,978 (27%)
Postgraduate	197 / 1,978 (10.0%)
Prefer not to say	6 / 1,978 (0.3%)
Vocational or Similar	7 / 1,978 (0.4%)

¹Mean (SD) [Min, Max]; n / N (%)

```
table(intake_participants$gender, useNA = "ifany")
```

Man	Non-binary	Prefer not to say	Woman
1282	105	3	587
<NA>			
1			

```
table(intake_participants$ethnicity, useNA = "ifany")
```

American Indian and Alaska Native	8
Asian	159
Black	145
Multiple	199
Native Hawaiian and Other Pacific Islander	2
Not reported	44
Other	53
Prefer not to say	2
White	1366

```
table(intake_participants$political_party, useNA = "ifany")
```

Conservative and Unionist Party	Democratic Party
30	591
Democratic Unionist Party (DUP)	Green Party of England and Wales
1	110
Independent Labour Party / Co-operative Party	
428	220
Liberal Democrats	No affiliation / Independent

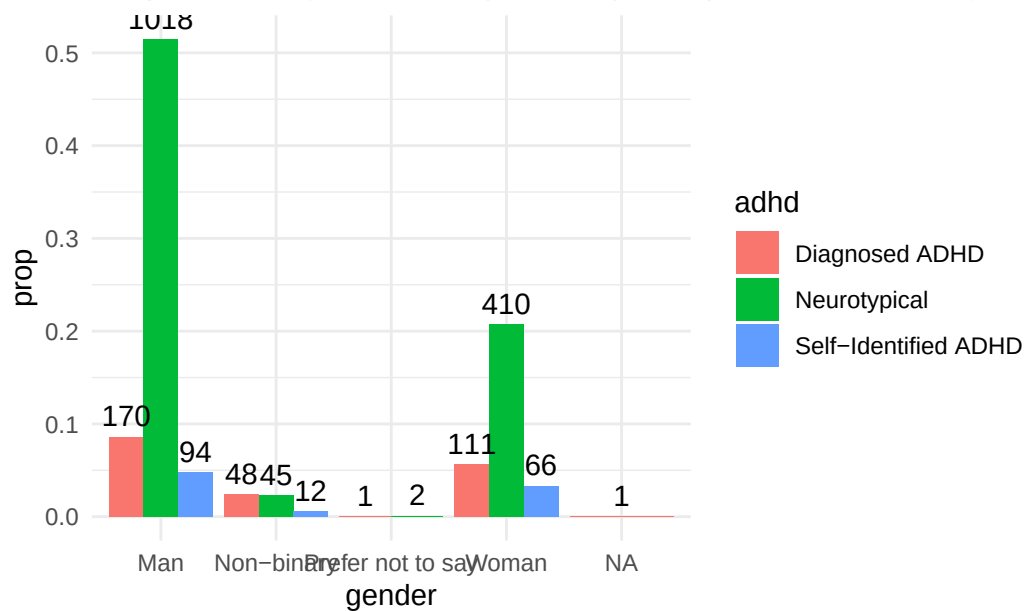
68	171
Other	Plaid Cymru
34	7
Prefer not to say	Reform UK
51	55
Republican Party	Scottish National Party (SNP)
178	31
Sinn Féin	
3	

```
table(intake_participants$edu_level, useNA = "ifany")
```

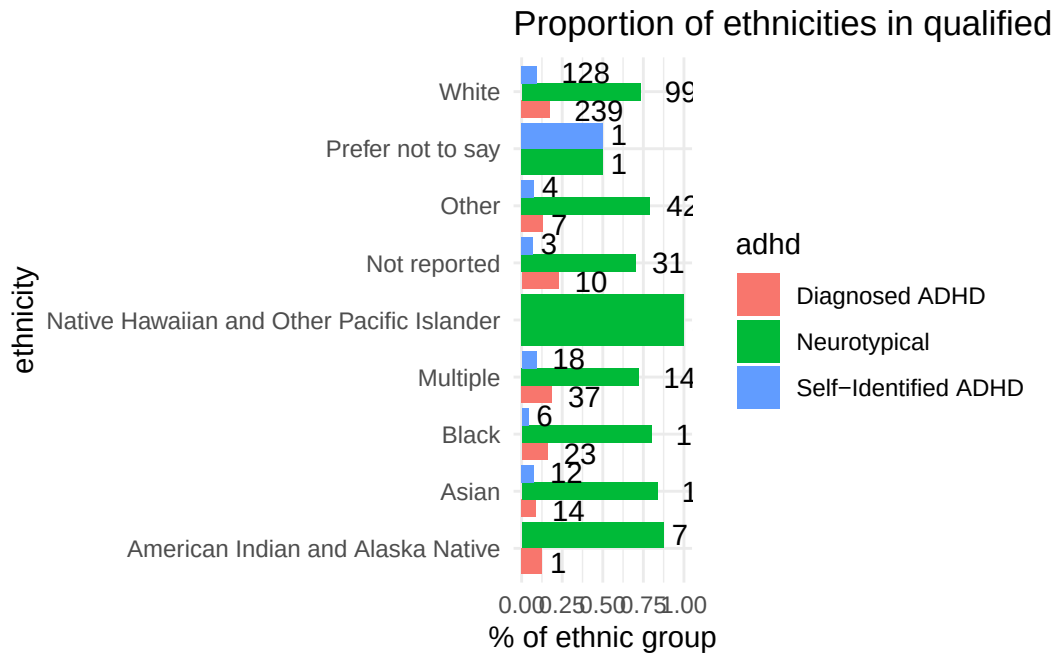
Associate degree	Bachelors degree
171	607
Below High School	High School
33	432
Higher education but no degree	Postgraduate
525	197
Prefer not to say	Vocational or Similar
6	7

```
# barplot of gender and ADHD
intake_participants |> count(gender, adhd) |> mutate(prop = n / sum(n)) |>
  ggplot(aes(x = gender, y = prop, fill = adhd)) +
  geom_col(position = "dodge") +
  geom_text(
    aes(label = n),
    position = position_dodge(width = 0.9),
    vjust = -0.5
  ) +
  labs(title = "Proportion of genders in qualified participants, coloured by ADHD indicator")
```

Proportion of genders in qualified participants, coloured by ADI



```
# barplot of ethnicity and adhd (proportion of ethnicity)
intake_participants |> count(ethnicity, adhd) |> group_by(ethnicity) |> mutate(prop = n / sum(n))
ggplot(aes(x = ethnicity, y = prop, fill = adhd)) +
  geom_col(position = "dodge") +
  geom_text(
    aes(label = n),
    position = position_dodge(width = 1),
    hjust = -0.5
  ) +
  coord_flip() +
  labs(title = "Proportion of ethnicities in qualified participants, by ADHD indicator", y = "proportion")
```



```
# barplot of video games platforms played
intake_participants_platforms <- intake_participants |>
  select(pid, starts_with("plays")) |>
  pivot_longer(
    cols = starts_with("plays_"),
    names_to = "platform",
    values_to = "play"
  ) |>
  mutate(
    play = as.integer(play),
    platform = str_replace(platform, "plays_", "")
  )

# pairplot of age, height, weight coloured by gender
intake_participants |> select(age, height, weight, gender) |>
  ggpairs(
    aes(colour = gender),
    columns = c("age", "height", "weight")
  )
```

Warning: Groups with fewer than two data points have been dropped.

Warning: Removed 1 row containing missing values or values outside the scale range

```
(`geom_density()`).
```

Warning: Removing 1 row that contained a missing value

Removing 1 row that contained a missing value

Warning: Groups with fewer than two data points have been dropped.

Warning: Removed 1 row containing missing values or values outside the scale range

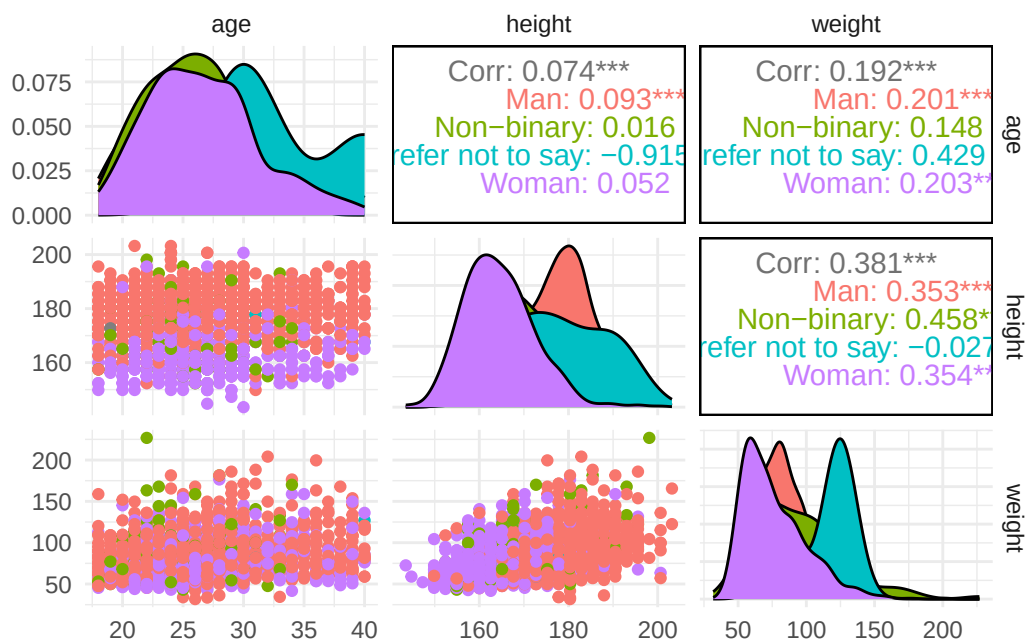
```
(`geom_density()`).
```

Warning: Removing 1 row that contained a missing value

Warning: Groups with fewer than two data points have been dropped.

Warning: Removed 1 row containing missing values or values outside the scale range

```
(`geom_density()`).
```



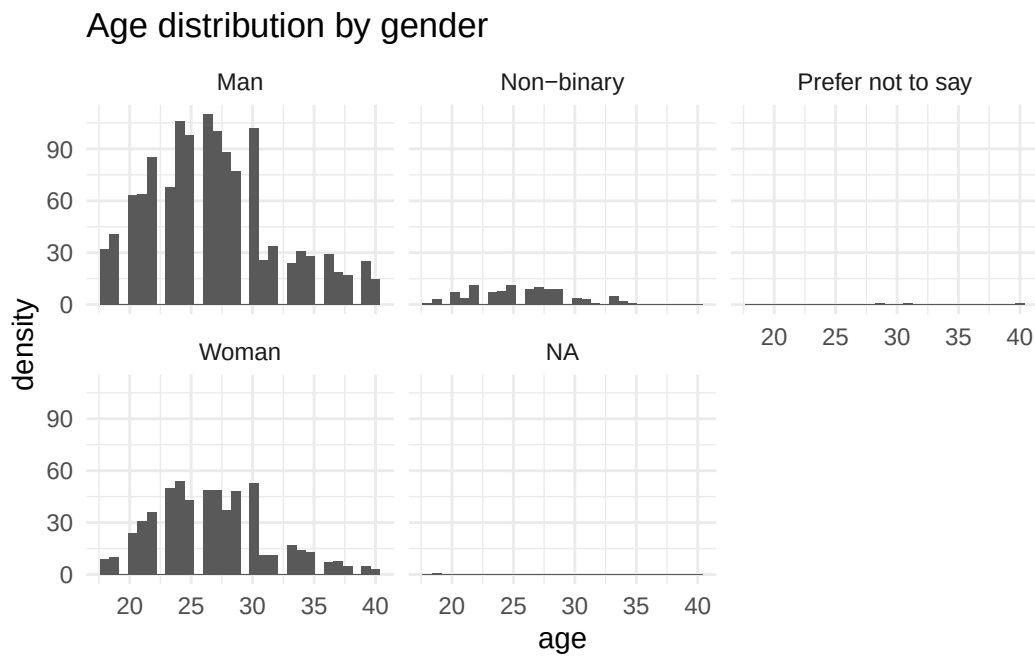
```
# distribution of age by gender
ggplot(intake_participants, aes(x = age)) +
  geom_histogram() +
  facet_wrap(vars(gender)) +
  labs(
```

```

    title = "Age distribution by gender",
    x = "age",
    y = "density",
  )

```

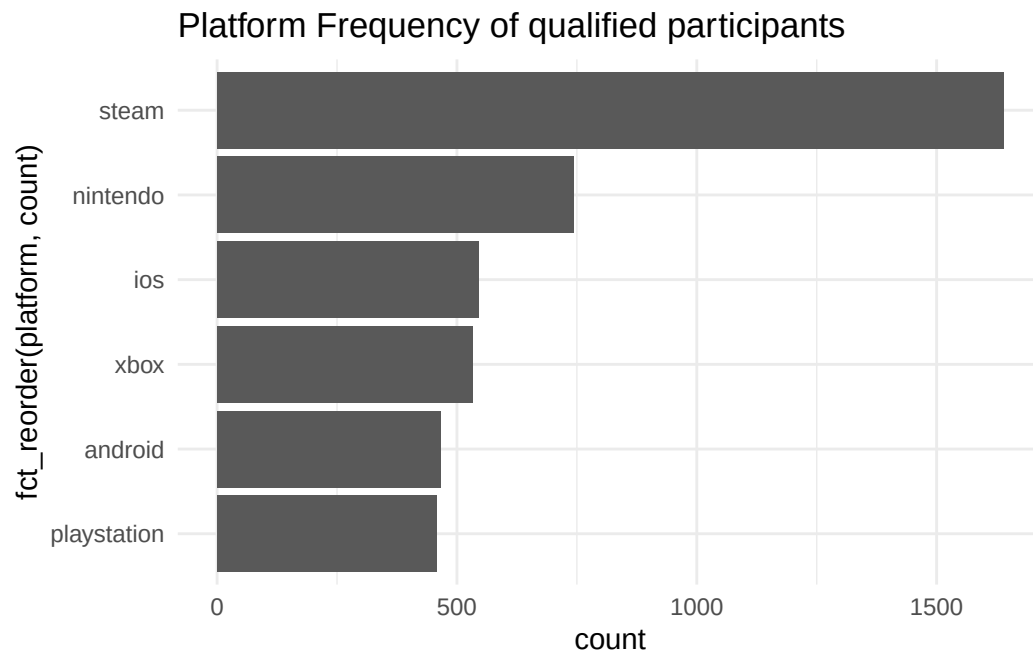
`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



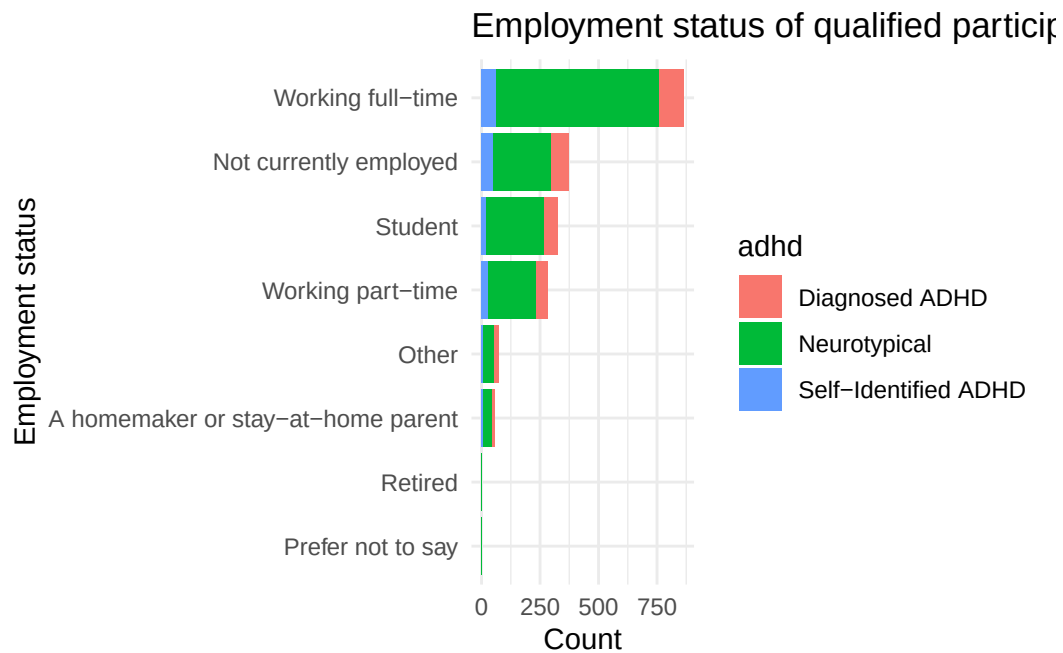
```

# distribution of platforms
intake_participants_platforms |>
  group_by(platform) |>
  summarise(count = sum(play)) |>
  ggplot(aes(x = fct_reorder(platform, count), y = count)) +
  geom_col() +
  labs(title = "Platform Frequency of qualified participants") +
  coord_flip()

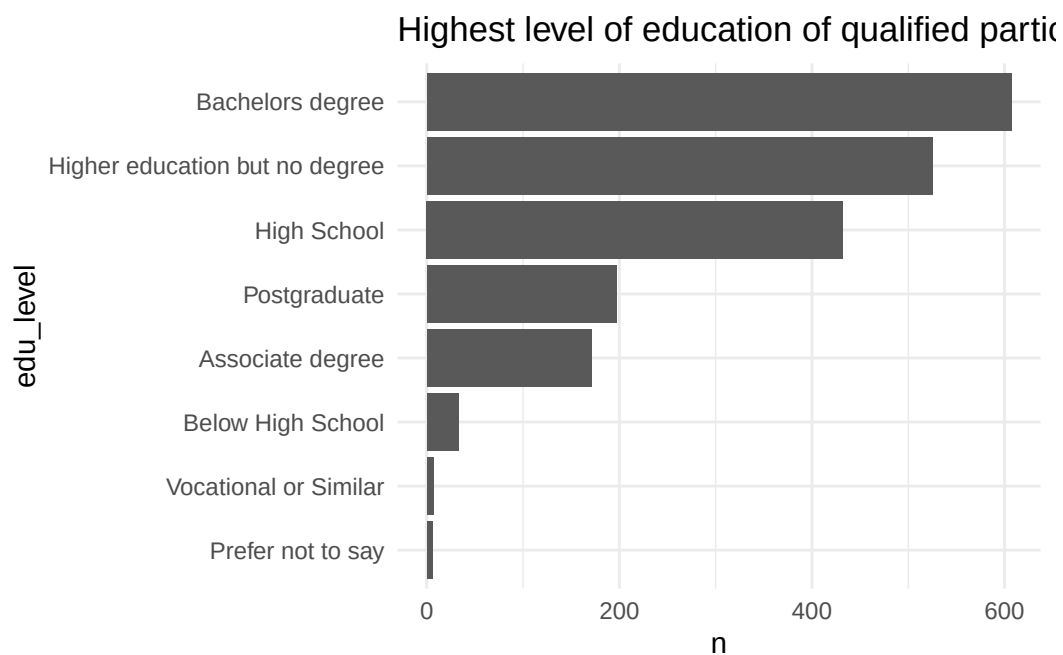
```



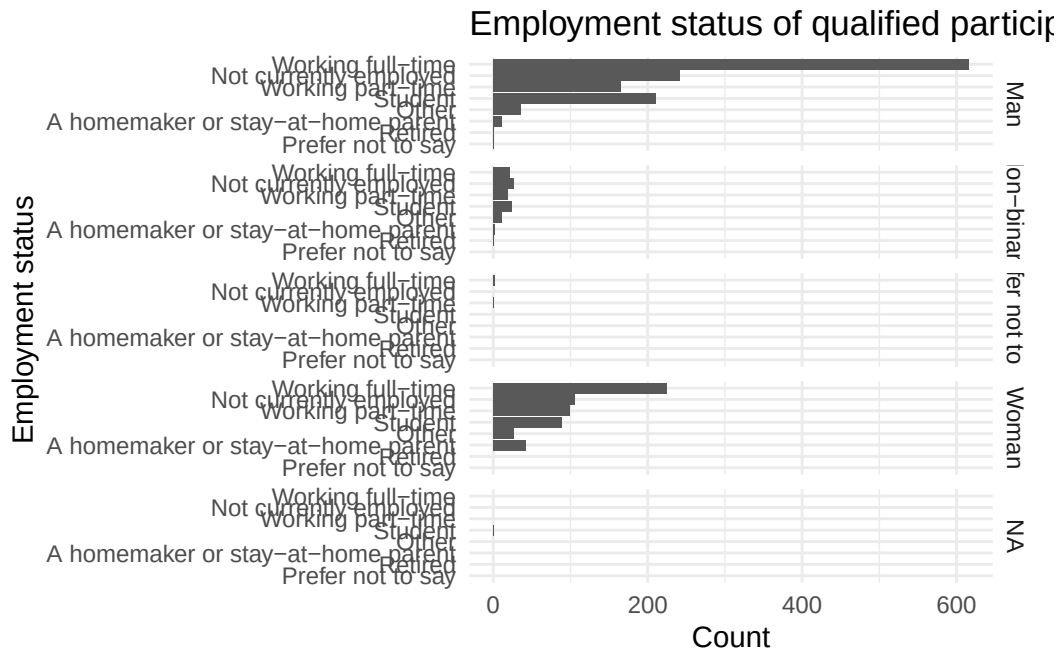
```
# employment status (with ADHD)
intake_participants |>
  filter(!is.na(employment)) |>
  count(employment, adhd) |>
  mutate(employment = fct_reorder(employment, n)) |>
  ggplot(aes(x = n, y = employment, fill = adhd)) +
  geom_col() +
  labs(
    x = "Count",
    y = "Employment status",
    title = "Employment status of qualified participants"
  )
```



```
# level of education
intake_participants |>
  count(edu_level) |>
  mutate(edu_level = fct_reorder(edu_level, n)) |>
  ggplot(aes(y = edu_level, x = n)) +
  geom_col() +
  labs(
    title = "Highest level of education of qualified participants"
  )
```



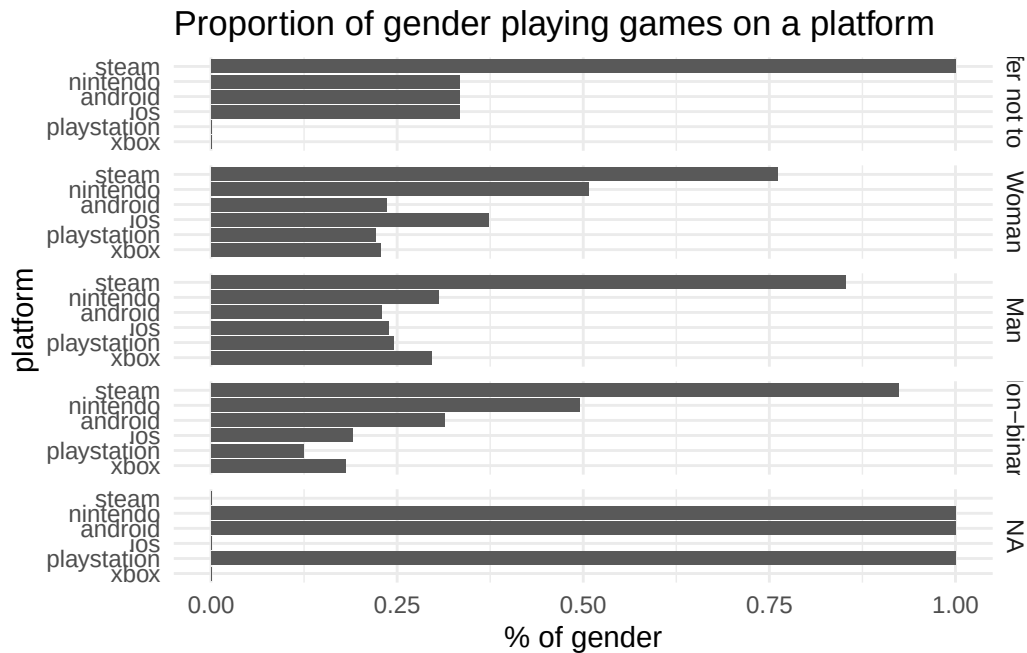
```
# employment status with gender
intake_participants |>
  filter(!is.na(employment)) |>
  count(employment, gender) |>
  mutate(employment = fct_reorder(employment, n)) |>
  ggplot(aes(x = n, y = employment)) +
  geom_col() +
  labs(
    x = "Count",
    y = "Employment status",
    title = "Employment status of qualified participants"
  ) +
  facet_grid(rows = vars(gender))
```

```
# trends between platform and adhd
intake_participants_platforms |>
  left_join(intake_participants |> select(pid, gender, adhd), by = join_by(pid)) |>
  group_by(gender, platform) |>
  summarise(count = sum(play), prop = count / n()) |>
  ggplot(aes(x = fct_reorder(platform, prop), y = prop)) +
  geom_col(position = position_dodge()) +
  facet_grid(rows = vars(fct_reorder(gender, prop, .desc = TRUE))) +
  labs(x = "platform", y = "% of gender", title = "Proportion of gender playing games on a p")
coord_flip()
```

`summarise()` has regrouped the output.

- i Summaries were computed grouped by gender and platform.
- i Output is grouped by gender.
- i Use `summarise(.groups = "drop\_last")` to silence this message.
- i Use `summarise(.by = c(gender, platform))` for per-operation grouping (`?dplyr::dplyr\_by`) instead.



We can see the following: a larger proportion of qualified women in the dataset have ADHD compared to men, a third of white participants have ADHD, a quarter of multiple ethnicities have ADHD, the Asian ethnic group had the smallest proportion of people with ADHD, steam is the most popular platform to play followed by Nintendo, nearly 50% of women played on their Nintendo device compared to roughly 30% of men.

Additionally, most of the qualified participants are in their 20s, working full time or unemployed.